

University of Verona

A.Y. 2025-26

Machine Learning & Deep Learning

Deep Learning

Beyond Supervised Learning

Self-Supervised Learning

Vittorio Murino

Credits

- Pietro Morerio
- Nuno C. Garcia

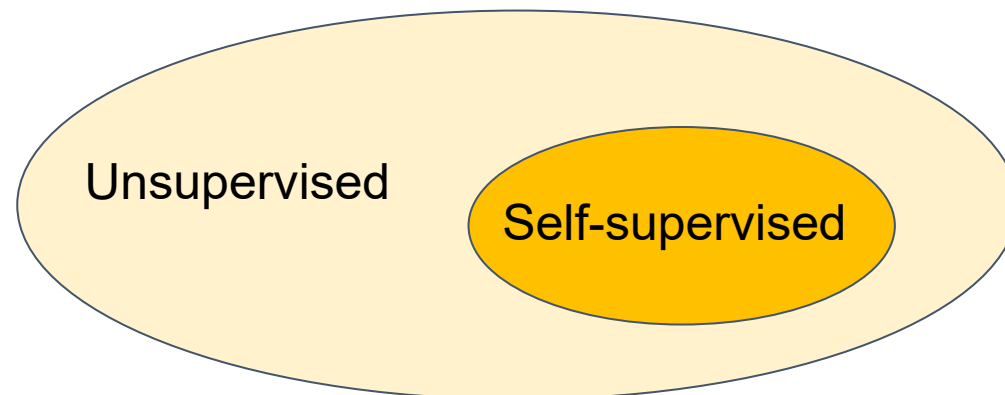
The Problem

1. DNNs need lots of data for generalizing well

➡ *Pretraining on a large dataset for a similar task usually helps (e.g. ImageNet)*

2. Obtaining human annotations for large datasets is expensive/unfeasible

➡ *Unsupervised representation learning (i.e. learning without human annotation)*



Why Self-Supervision?

- It's expensive to build custom datasets for all tasks
- Almost impossible in some areas, e.g. medical data
- Huge number of unlabelled images and videos – Facebook, YouTube, etc.
- How children learn

Why Self-Supervision?



Vittorio Murino

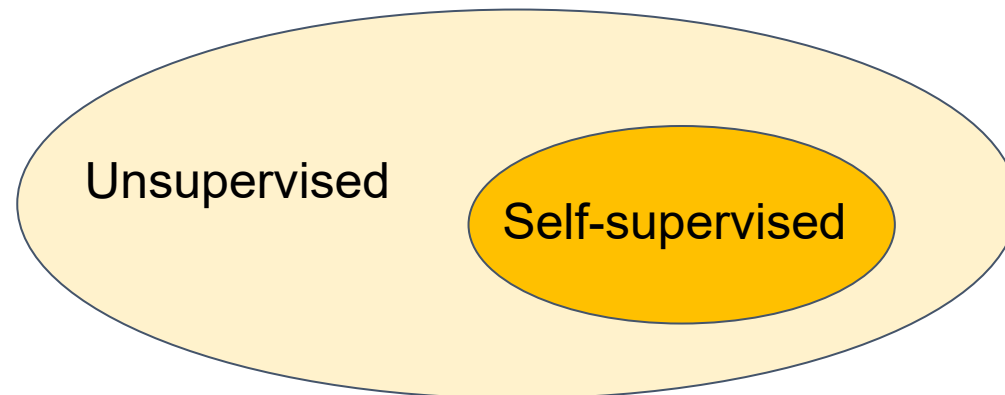
What is Self Supervision?

1. DNNs need lots of data for generalizing well

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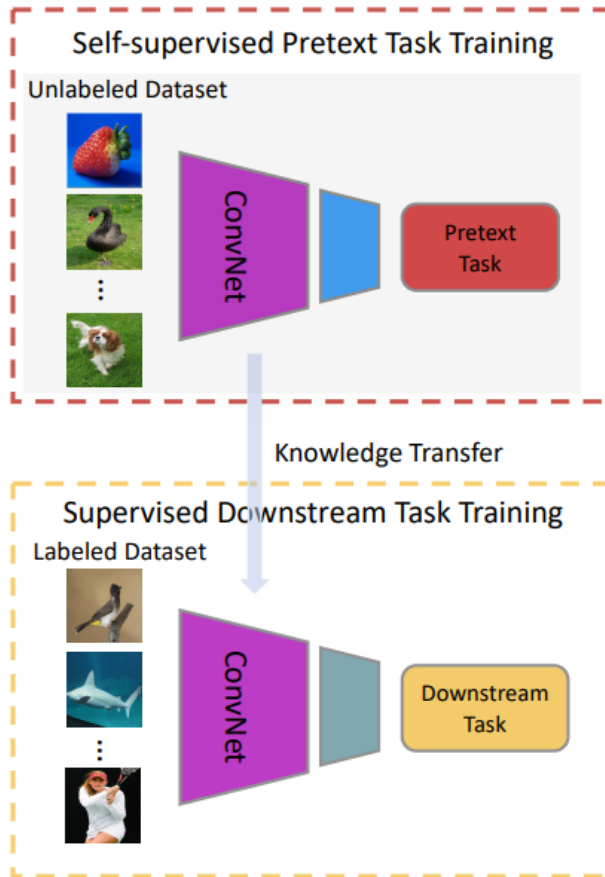
➡ Find a **pretext task** where the source of **supervision** lies **in the data itself**



What is Self Supervision?

- Data provides the supervision signal.
- General:
 - hide part of the data;
 - get a network to predict the hidden part given the available part.
- Define an auxiliary task that, in order to be solved, needs some semantic representation.

Tasks [1]



- **Pretext task(s):**
 - Pre-designed task to learn features
 - Lots of data
 - No human annotation
- **Downstream task(s):**
 - Main task(s) to be solved
 - Possibly few data
 - Human annotation needed
 - Also used to evaluate the quality of features learned by the self-supervised pretext task
 - Can be the same task

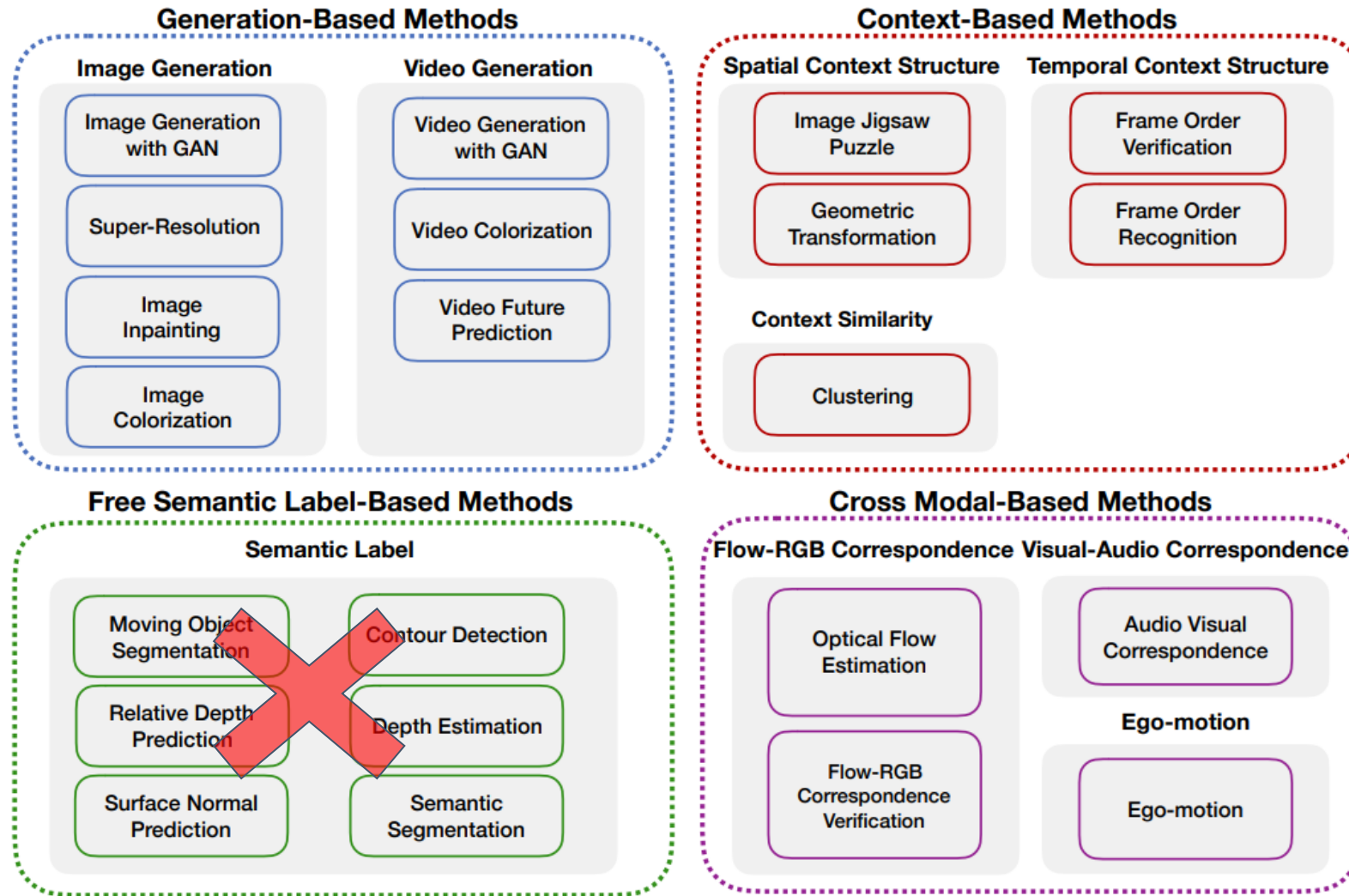
[1] Longlong Jing, Yingli Tian, *Self-supervised Visual Feature Learning with Deep Neural Networks: A Survey*, <https://arxiv.org/abs/1902.06162>, 2019

Designing pretext tasks

- Supervision cannot come from humans
 - Must come from the data itself
 - Can be automatically generated by other algorithms [1] (arguable in my opinion)
- The pretext task must involve **visual reasoning** in order to force the DNNs to learn (good) visual and/or semantic features.

[1] Longlong Jing, Yingli Tian, *Self-supervised Visual Feature Learning with Deep Neural Networks: A Survey*, <https://arxiv.org/abs/1902.06162>

Pretext task categories [1]



[1] Longlong Jing, Yingli Tian, *Self-supervised Visual Feature Learning with Deep Neural Networks: A Survey*, <https://arxiv.org/abs/1902.06162>

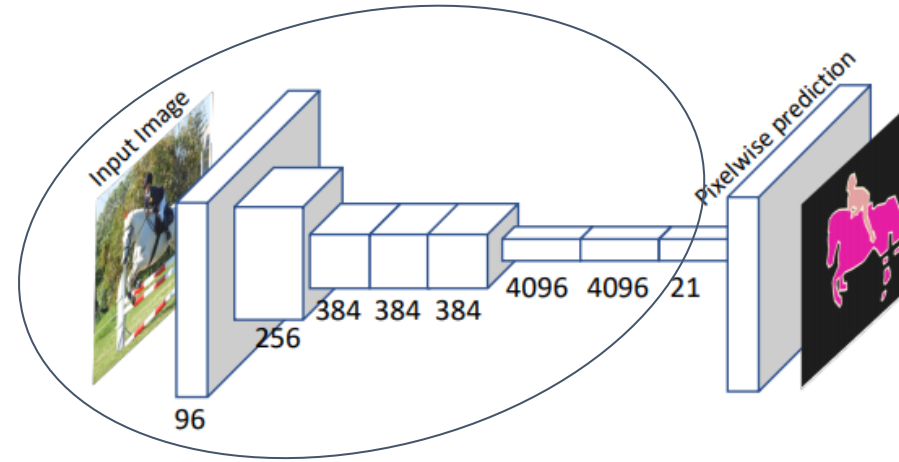
Downstream tasks

- Images

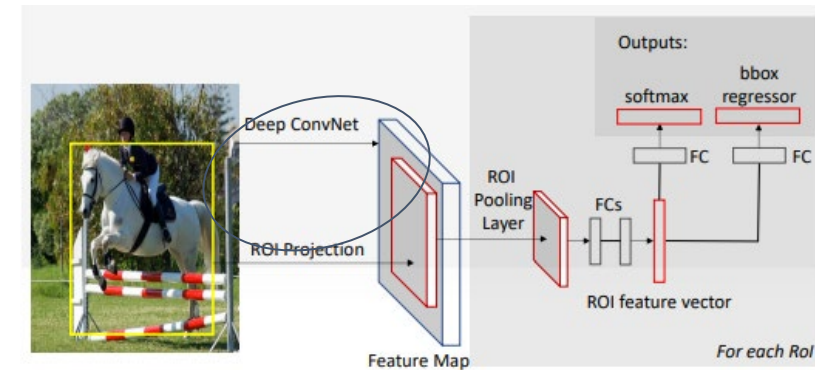
- Image classification
- Semantic segmentation
- Object detection
- ...

- Videos

- Human action recognition
- ...



FCN [2] for semantic segmentation



Fast R-CNN [3] for object detection

[2] J. Long, E. Shelhamer, and T. Darrell, *Fully convolutional networks for semantic segmentation*, CVPR 2015

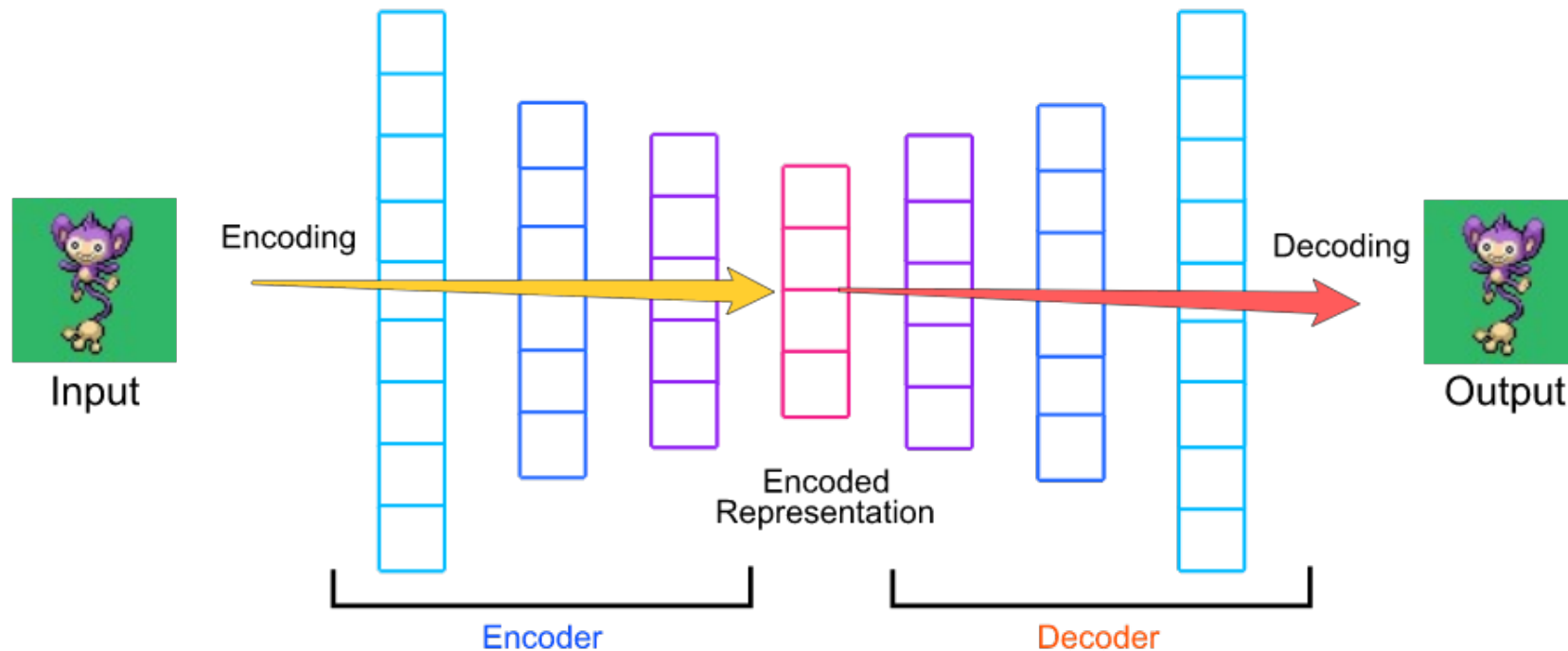
[3] S. Ren, K. He, R. Girshick, and J. Sun, *Faster r-cnn: Towards realtime object detection with region proposal networks*, in NIPS 2015

Using images

Generation-based pretext task

Autoencoder

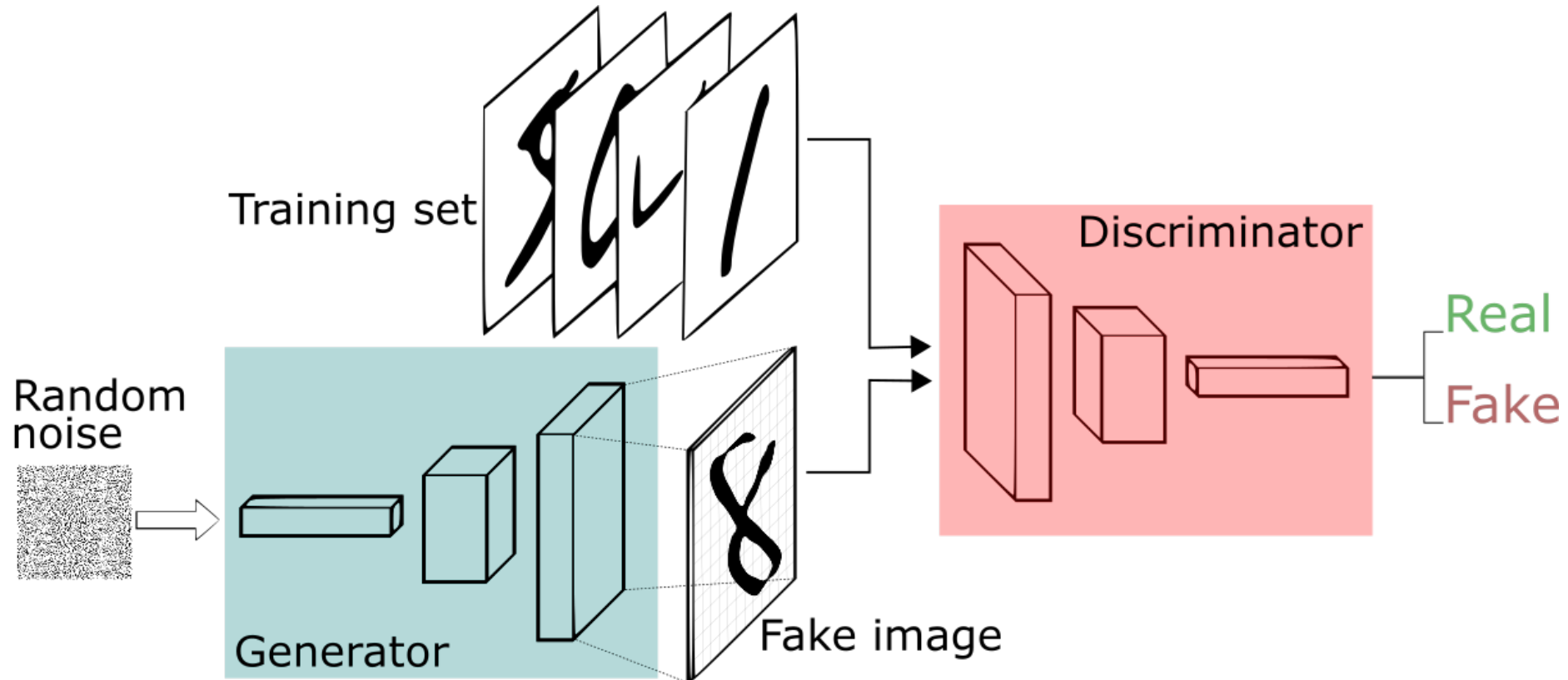
Generation-based pretext tasks



G.E. Hinton and R.R. Salakhutdinov, "Reducing the dimensionality of data with neural networks," *Science*, 2006

GAN – Generative Adversarial Networks

Generation-based pretext tasks

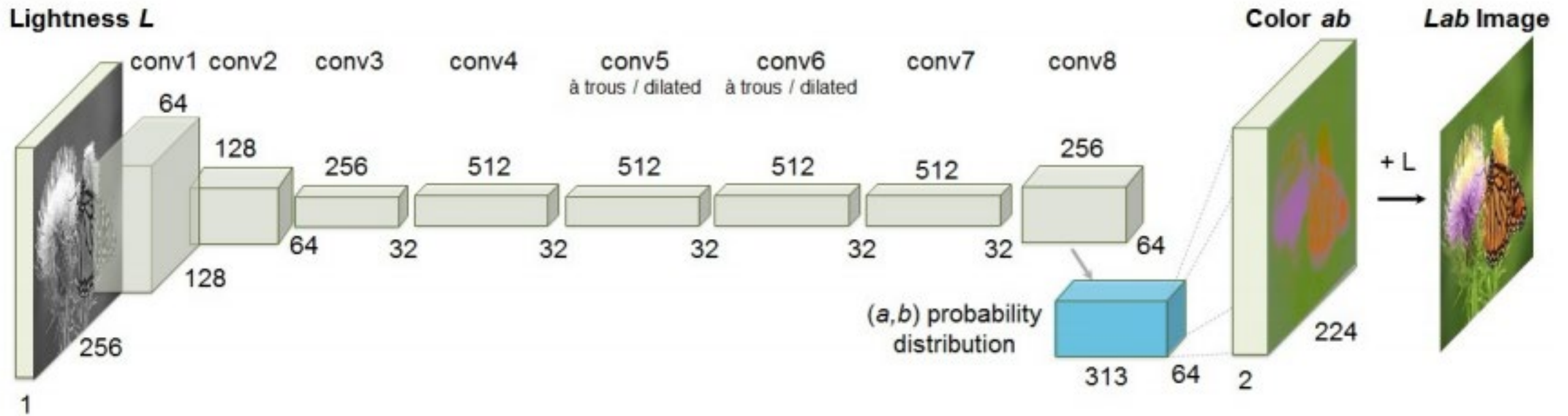


[5] I. Goodfellow, J. Pouget-Abadie, M. Mirza, B. Xu, D. WardeFarley, S. Ozair, A. Courville, Y. Bengio, “Generative adversarial nets”, *NIPS 2014*.

Image colorization

Generation-based pretext tasks

- Train network to predict pixel color from a monochrome input
- Classification with class re-balancing based on ImageNet statistics

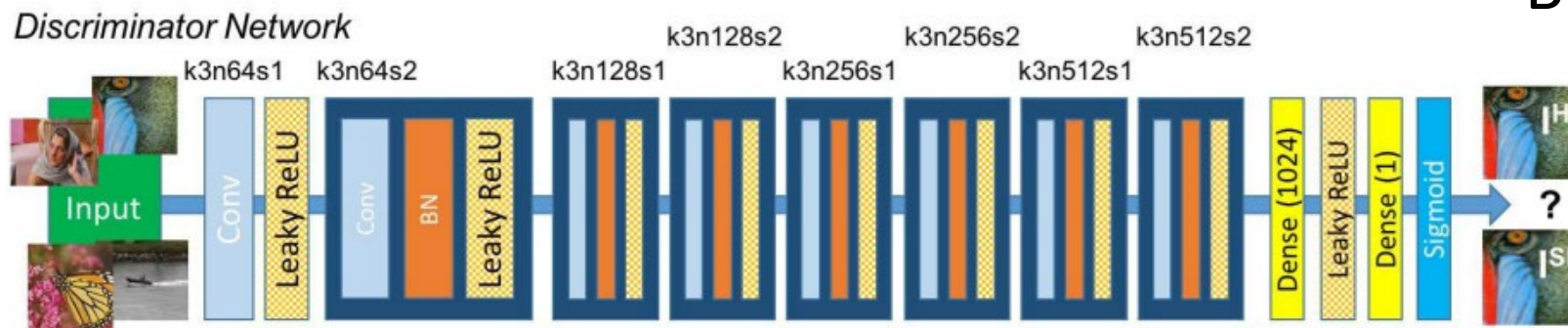
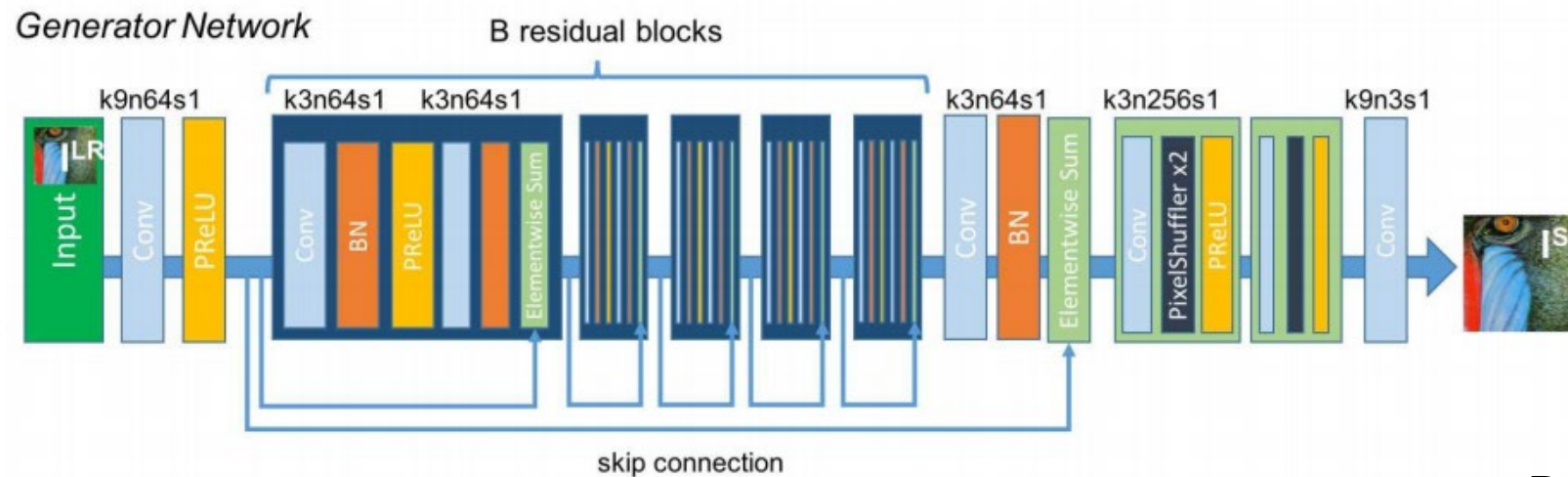


[8] R. Zhang, P. Isola, and A. A. Efros, "Colorful image colorization," *ECCV 2016*.

SRGAN - Super resolution GAN

Generation-based pretext tasks

L2 + adversarial loss



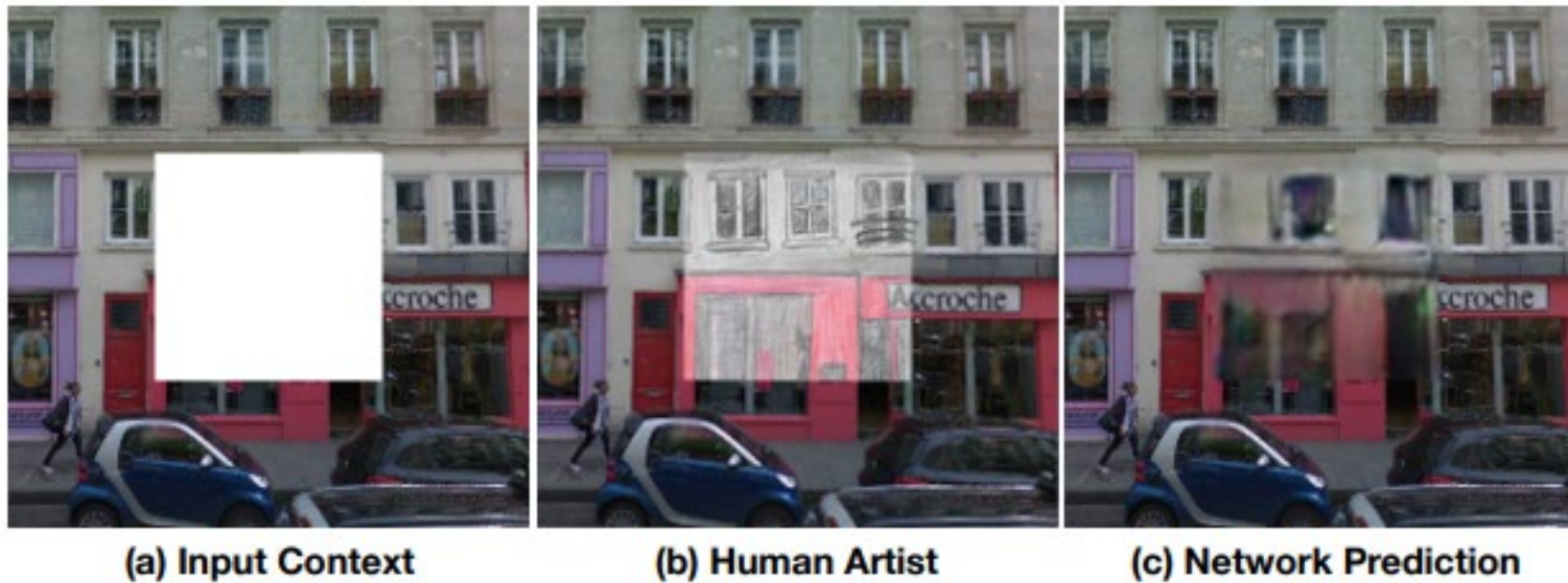
Binary CE loss

[6] C. Ledig et al., "Photo-realistic single image super-resolution using a generative adversarial network," *CVPR 2017*.

Image inpainting

Generation-based pretext tasks

L2 + adversarial loss



[7] D. Pathak, P. Krahenbuhl, J. Donahue, T. Darrell, A. Efros, “Context encoders: Feature learning by inpainting”, *CVPR 2016*

Video

Generation-based pretext tasks

- Video generation with GANs - VideoGAN [9]
- Video colorization [10]
- Future frame generation [11]

[9] C. Vondrick, H. Pirsiavash, and A. Torralba, “Generating videos with scene dynamics,” *NIPS 2016*.

[10] D. Tran, L. Bourdev, R. Fergus, L. Torresani, and M. Paluri, “Deep end2end voxel2voxel prediction,” *CVPRW 2016*.

[11] R. Villegas, J. Yang, S. Hong, X. Lin, and H. Lee, “Decomposing motion and content for natural video sequence prediction,” *ICLR 2017*.

Using images

Context-based pretext task

Geometric transformation: Image Rotations

Context-based pretext tasks

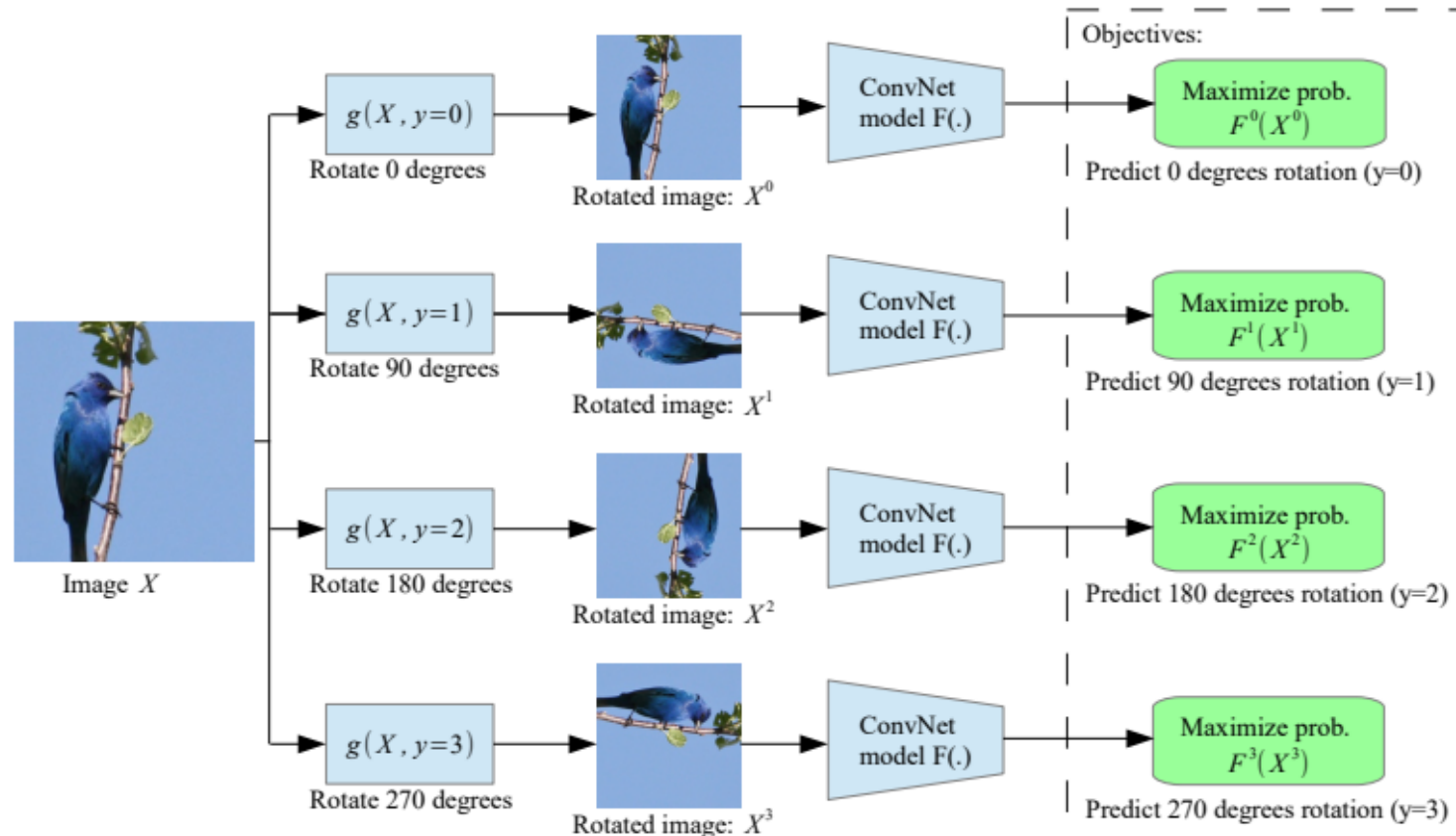
Which image has the correct rotation?



S. Gidaris, P. Singh, N. Komodakis, "Unsupervised representation learning by predicting image rotations," *ICLR 2018*.

Geometric transformation

Context-based pretext tasks



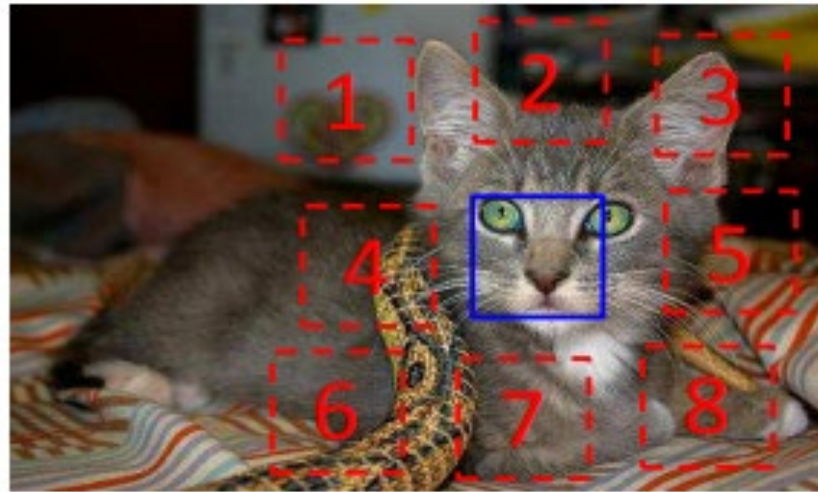
S. Gidaris, P. Singh, N. Komodakis, "Unsupervised representation learning by predicting image rotations," *ICLR 2018*.

Image Rotations

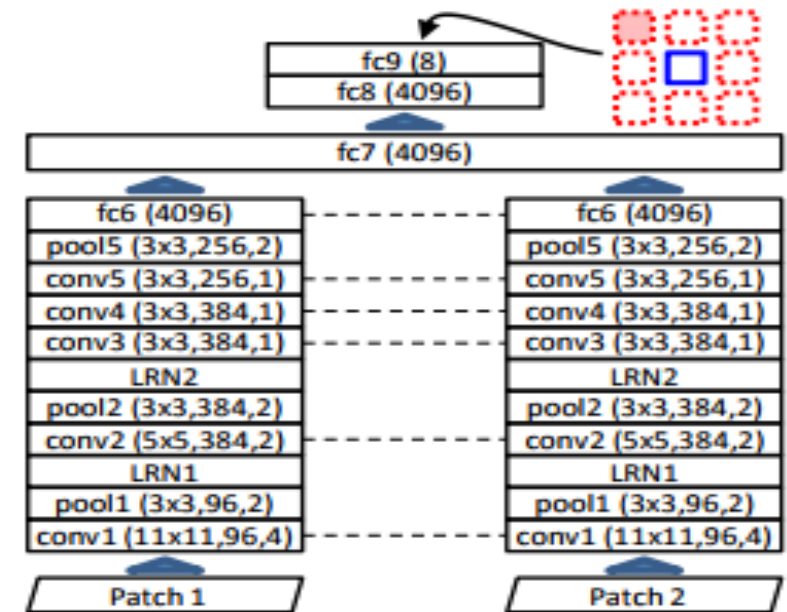
	Classification (%mAP)		Detection (%mAP)	Segmentation (%mIoU)
Trained layers	fc6-8	all	all	all
ImageNet labels	78.9	79.9	56.8	48.0
Random		53.3	43.4	19.8
Random rescaled Krähenbühl et al. (2015)	39.2	56.6	45.6	32.6
Egomotion (Agrawal et al., 2015)	31.0	54.2	43.9	
Context Encoders (Pathak et al., 2016b)	34.6	56.5	44.5	29.7
Tracking (Wang & Gupta, 2015)	55.6	63.1	47.4	
Context (Doersch et al., 2015)	55.1	65.3	51.1	
Colorization (Zhang et al., 2016a)	61.5	65.6	46.9	35.6
BIGAN (Donahue et al., 2016)	52.3	60.1	46.9	34.9
Jigsaw Puzzles (Noroozi & Favaro, 2016)	-	67.6	53.2	37.6
NAT (Bojanowski & Joulin, 2017)	56.7	65.3	49.4	
Split-Brain (Zhang et al., 2016b)	63.0	67.1	46.7	36.0
ColorProxy (Larsson et al., 2017)		65.9		38.4
Counting (Noroozi et al., 2017)	-	67.7	51.4	36.6
(Ours) RotNet	70.87	72.97	54.4	39.1

Table 7: **Task & Dataset Generalization: PASCAL VOC 2007 classification and detection results, and PASCAL VOC 2012 segmentation results.** We used the publicly available testing frameworks of Krähenbühl et al. (2015) for classification, of Girshick (2015) for detection, and of Long et al. (2015) for segmentation. For classification, we either fix the features before conv5 (column *fc6-8*) or we fine-tune the whole model (column *all*). For detection we use multi-scale training and single scale testing. All approaches use AlexNet variants and were pre-trained on ImageNet without labels except the ImageNet labels and Random entries. After unsupervised training, we absorb the batch normalization units on the linear layers and we use the weight rescaling technique proposed by Krähenbühl et al. (2015) (which is common among the unsupervised methods). As customary, we report the mean average precision (mAP) on the classification and detection tasks, and the mean intersection over union (mIoU) on the segmentation task.

Relative Position task



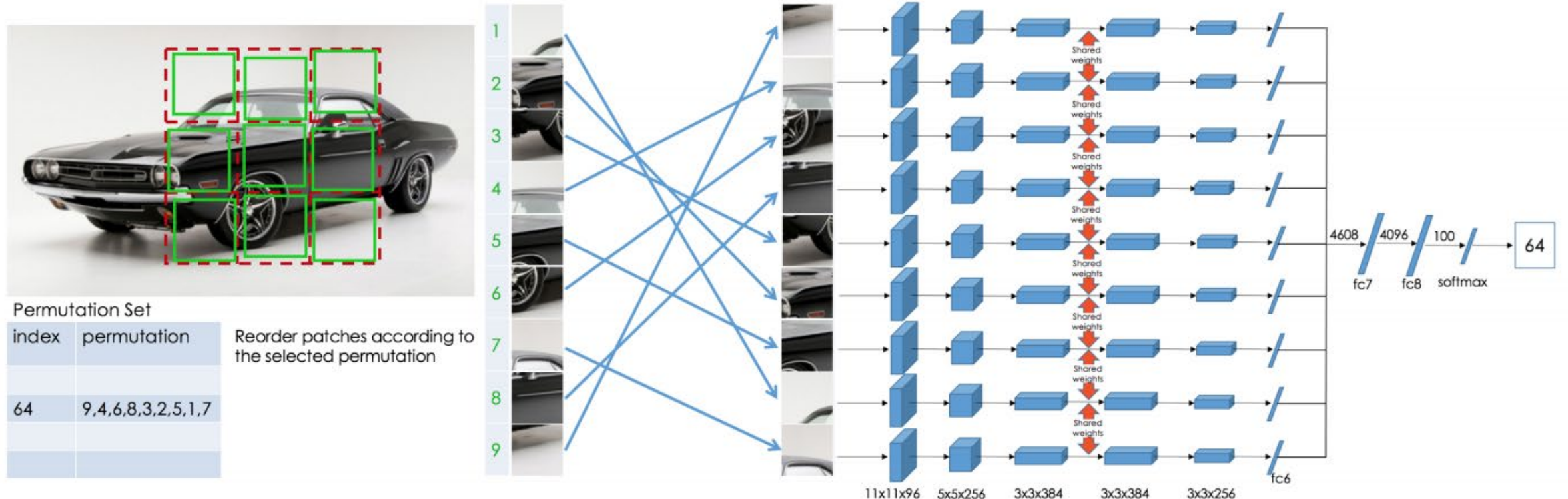
$$X = (\text{cat face patch}, \text{cat ear patch}); Y = 3$$



C. Doersch, A. Gupta, A. Efros, "Unsupervised visual representation learning by context prediction," *ICCV 2015*.

Solving a Jigsaw Puzzle

Context-based pretext tasks

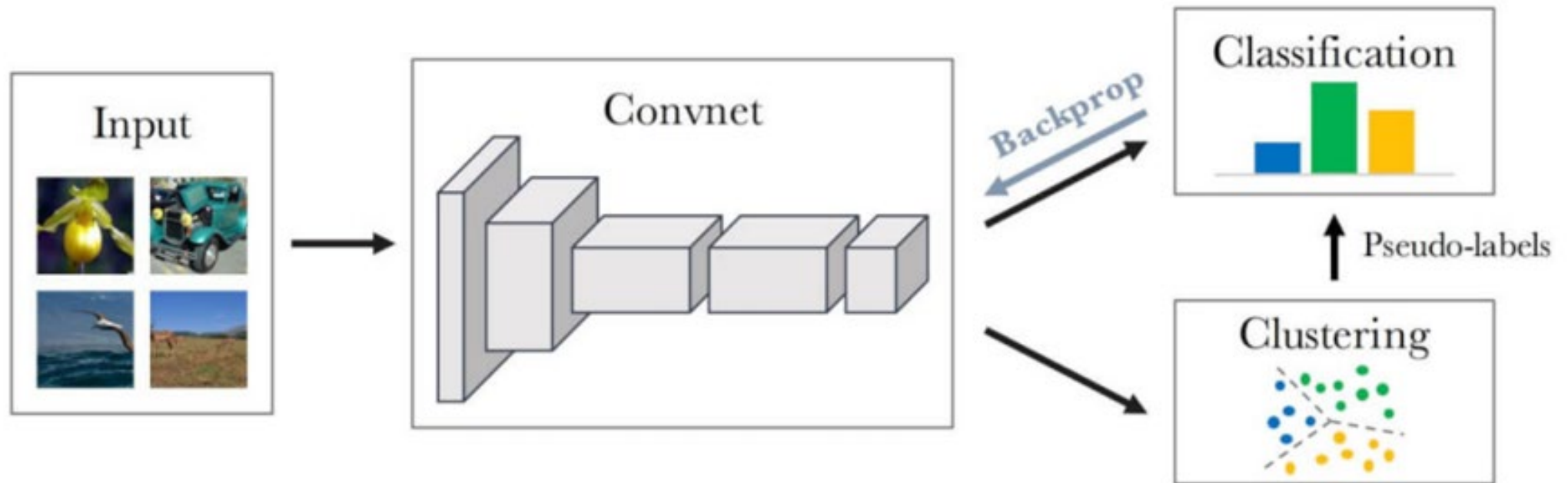


M. Noroozi and P. Favaro, "Unsupervised learning of visual representations by solving jigsaw puzzles," *ECCV 2016*.

Clustering

Context-based pretext tasks

- Iterative



M. Caron, P. Bojanowski, A. Joulin, M. Douze, “Deep clustering for unsupervised learning of visual features,” *ECCV 2018*.

Multi-task Learning

Findings:

- Deeper network improves performance (ResNet vs AlexNet)
- Colour and Rel-Pos superior to Exemplar
- Gap between self-supervision and strong supervision closing

Self-supervision task	ImageNet Classification top-5 accuracy	PASCAL VOC Detection mAP
Rel. Pos	59.21	66.75
Colour	62.48	65.47
Exemplar	53.08	60.94
Rel. Pos + colour	66.64	68.75
Rel. Pos + Exemplar	65.24	69.44
Rel. Pos + colour + Exemplar	68.65	69.48
ImageNet labels	85.10	74.17

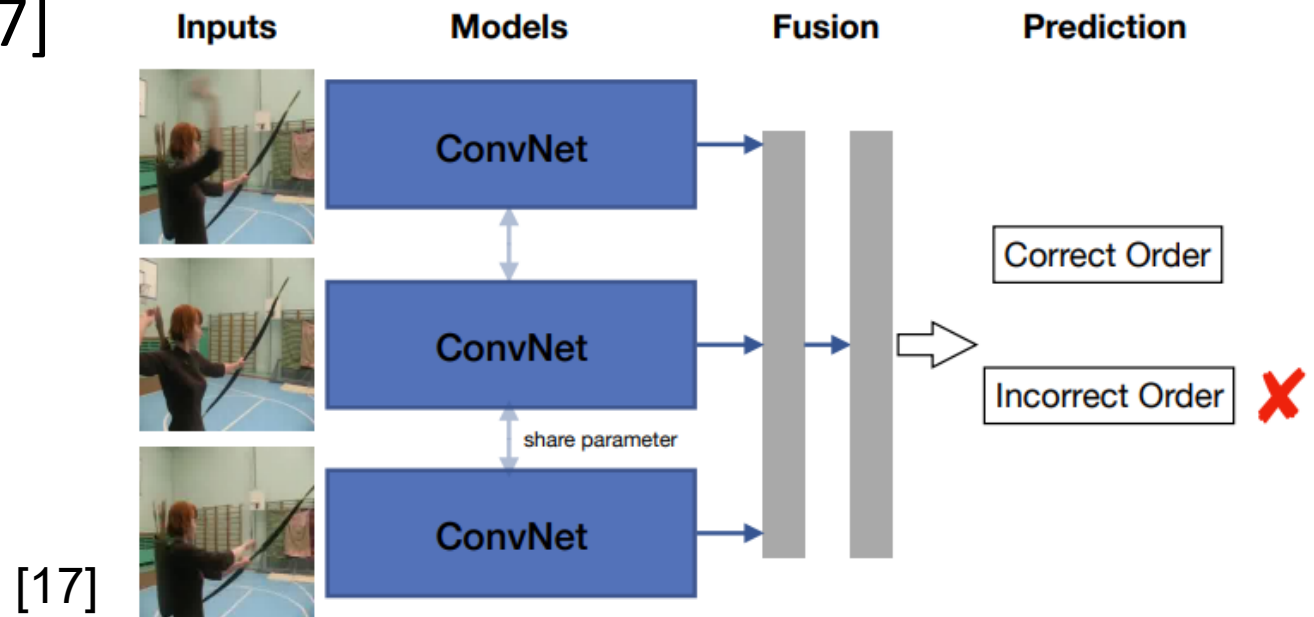
Using videos

Context-based pretext tasks

Video

Context-based pretext tasks

- Frame order recognition [16]
- Frame order verification [17]



[16] H.-Y. Lee, J.-B. Huang, M. Singh, M.-H. Yang, "Unsupervised representation learning by sorting sequences", *ICCV 2017*

[17] I. Misra, C. L. Zitnick, M. Hebert, "Shuffle and learn: unsupervised learning using temporal order verification", *ECCV 2016*

Learning and Using the Arrow of Time



Figure 1: Seeing these ordered frames from videos, can you tell whether each video is playing forward or backward? (answer below¹). Depending on the video, solving the task may require (a) low-level understanding (e.g. physics), (b) high-level reasoning (e.g. semantics), or (c) familiarity with very subtle effects or with (d) camera conventions. In this work, we learn and exploit several types of knowledge to predict the arrow of time automatically with neural network models trained on large-scale video datasets.

Learning and Using the Arrow of Time

- First, can we train a reliable arrow of time classifier from large-scale natural videos while avoiding artificial cues (i.e. cues introduced during video production, not from the visual world)
- Second, what does the model learn about the visual world in order to solve this task
- Third, and last, can we apply such learned commonsense knowledge to other video analysis tasks?
 - video representation learning and video forensics

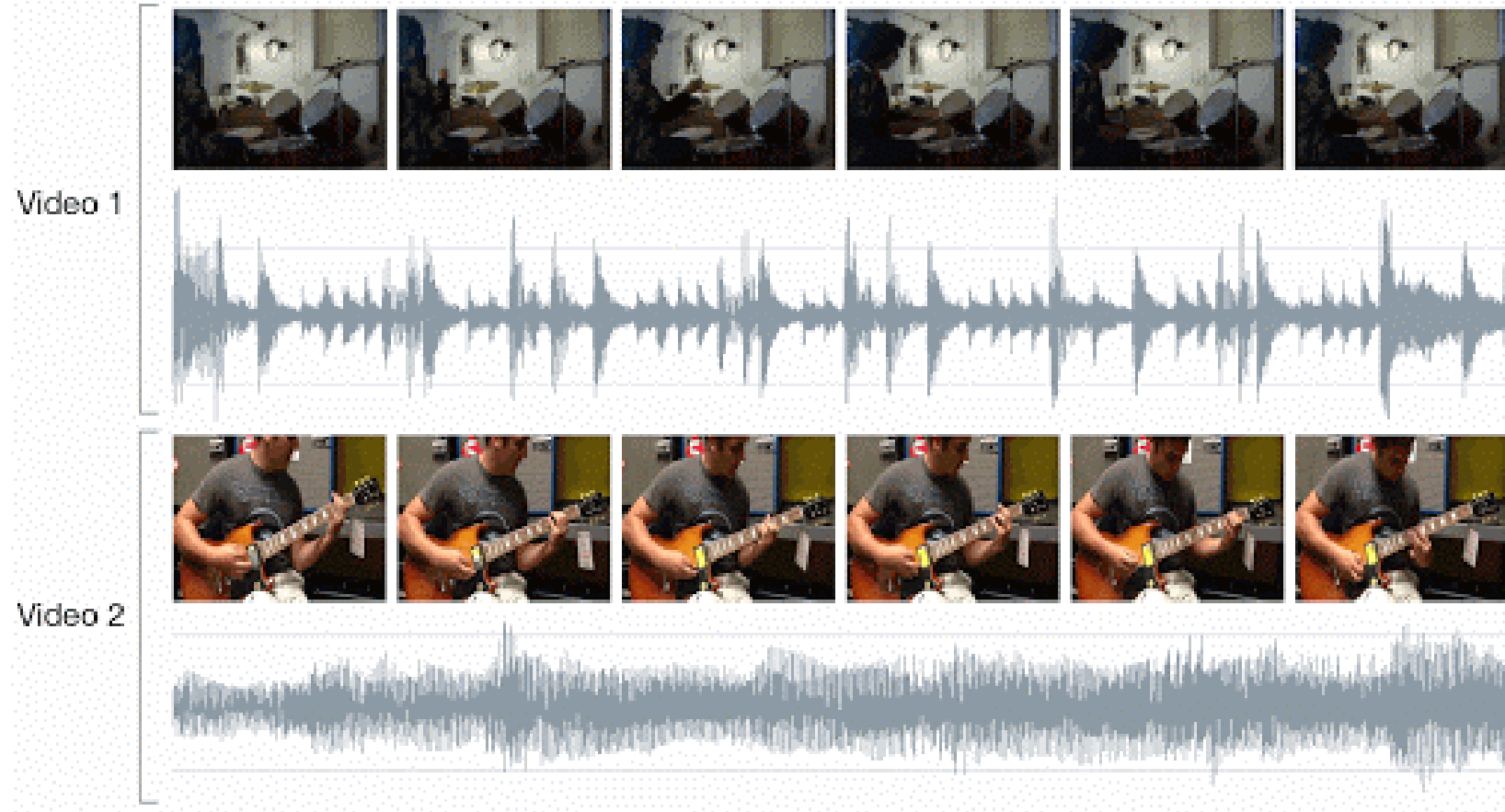
Sound + Vision

Multimodal pretext task

What can be learnt by watching and listening to videos?

- Good representations
 - Visual features
 - Audio features
- Intra- and cross-modal retrieval
 - Aligned audio and visual embeddings
- “What is making the sound?”
 - Learn to localize objects that sound

Sound + Vision



R. Arandjelovic and A. Zisserman, "Look, Listen, and Learn." ICCV 2017

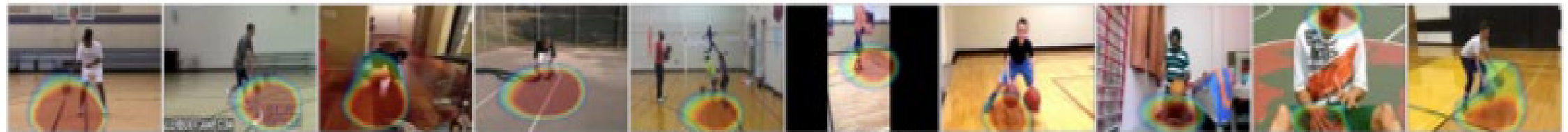
R. Arandjelovic and A. Zisserman, "Objects that sound", ECCV 2018

Audio-visual correspondence

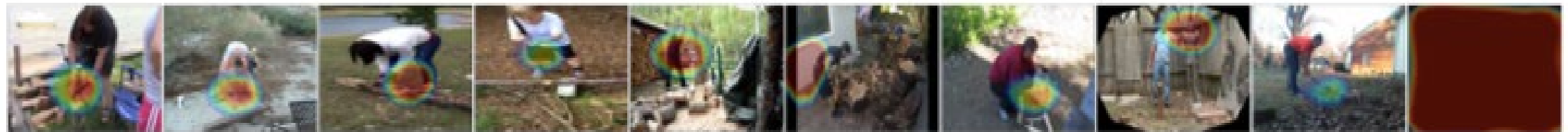
Multimodal pretext tasks

Two types of proxy task, different profit:

1. Predict audio-visual correspondence → semantics
2. Predict audio-visual synchronization → attention

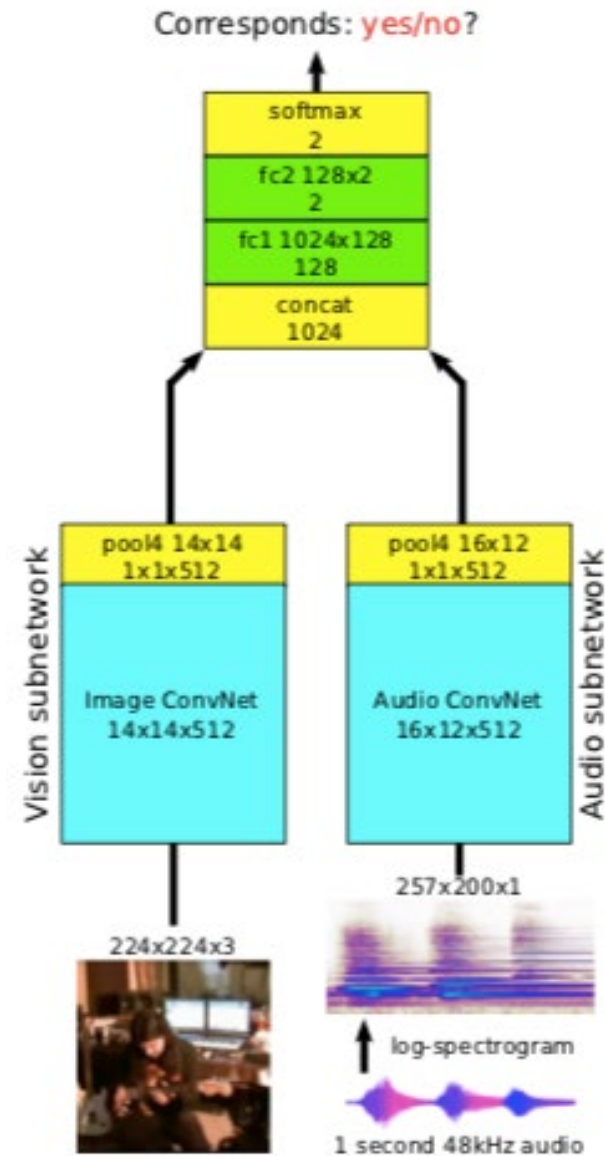


Dribbling basketball

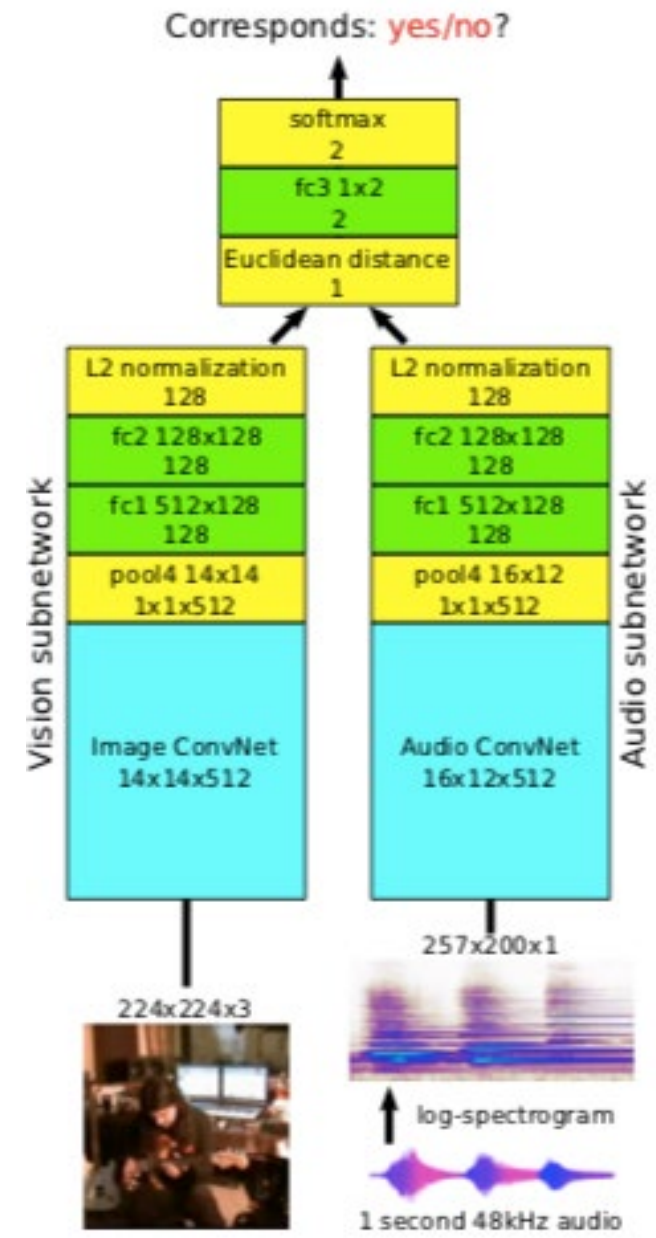


Chopping wood

Audio-visual correspondence



(d) L^3 -Net [4]



(c) AVE-Net

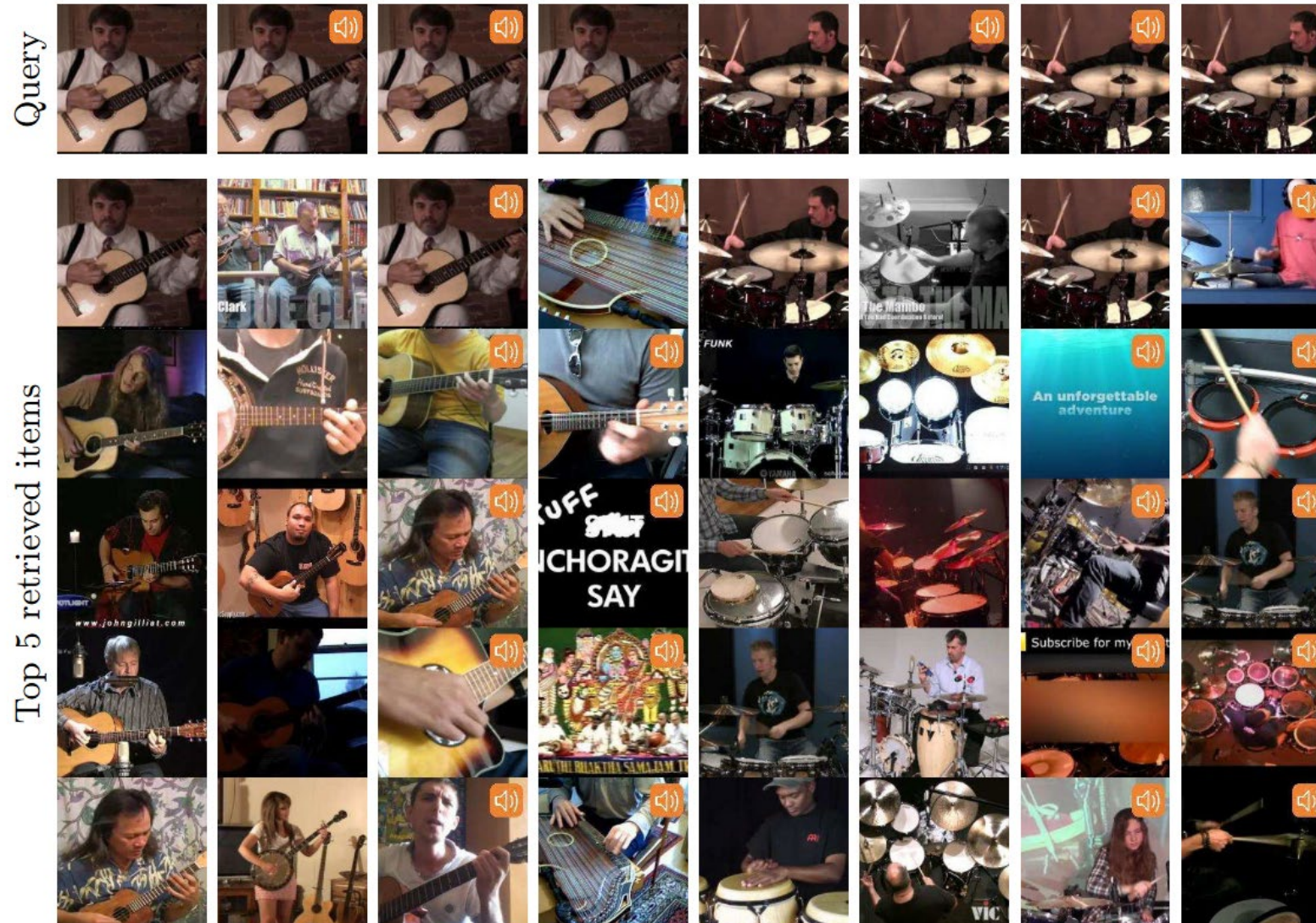
[4] R. Arandjelovic and A. Zisserman,
“Look, listen and learn”, ICCV 2017.

Retrieval performance

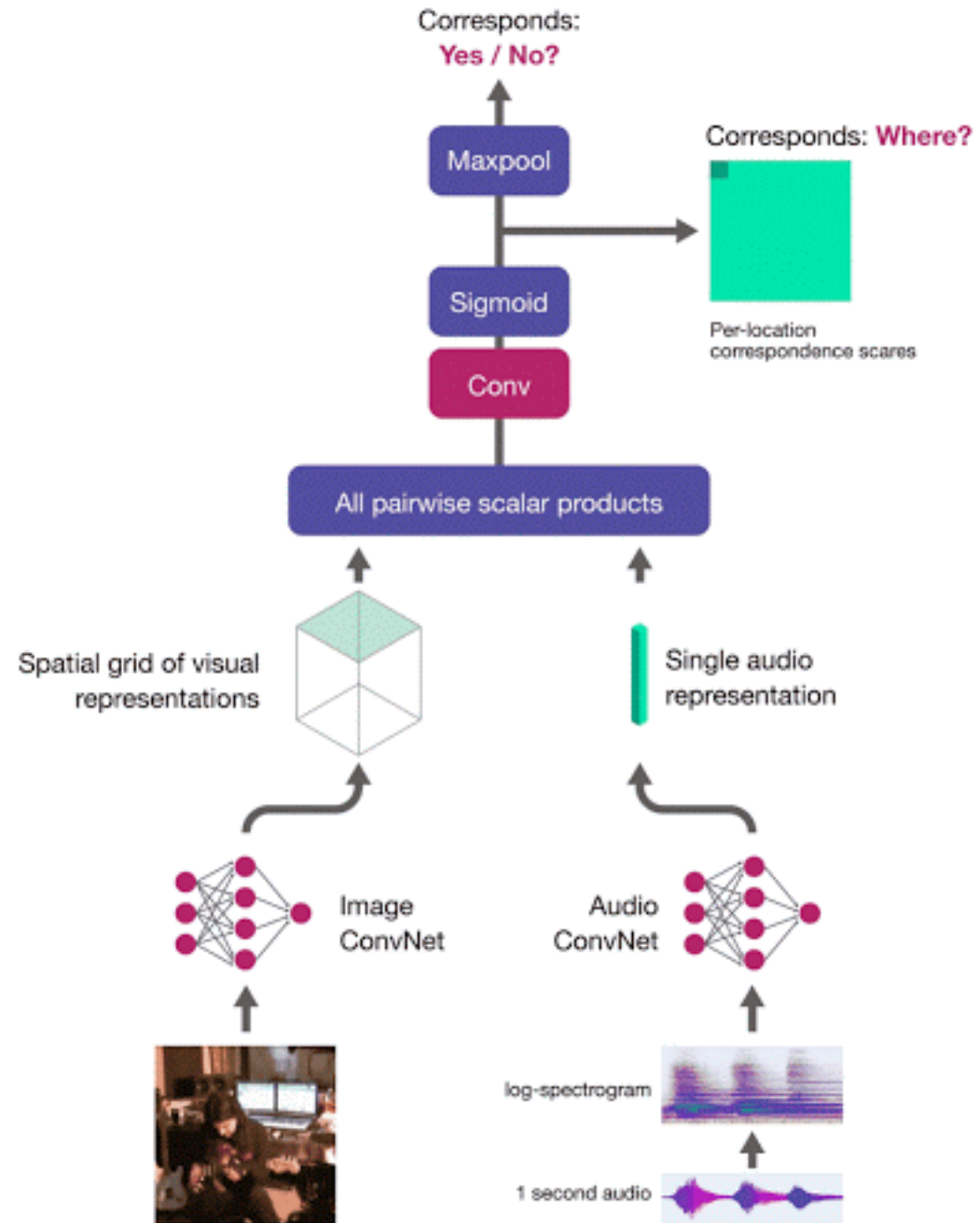
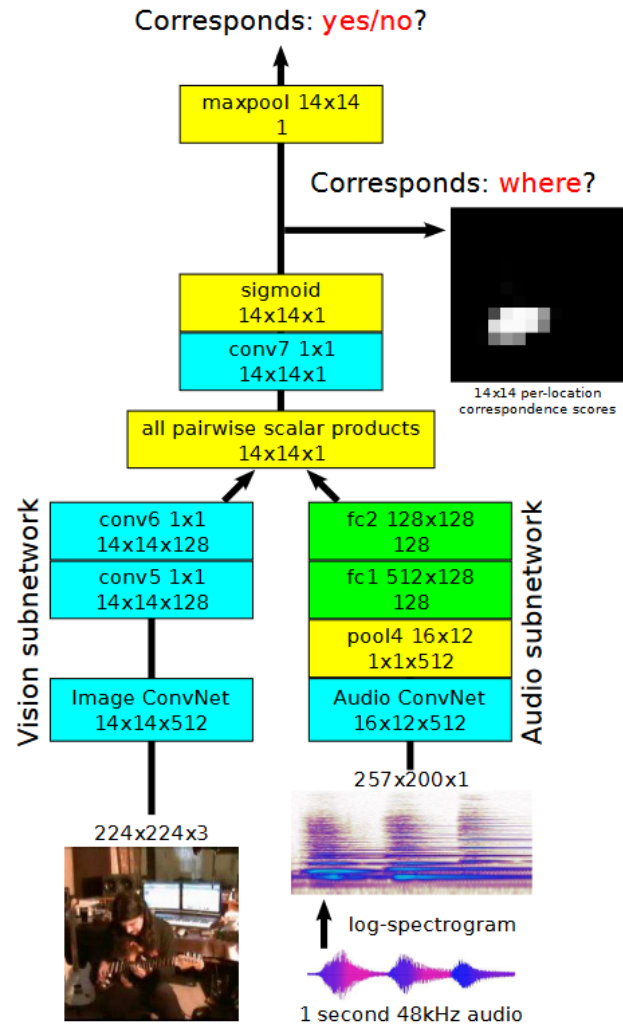
Method	im-im	im-aud	aud-im	aud-aud
Random chance	.407	.407	.407	.407
L^3 -Net [4]	.567	.418	.385	.653
L^3 -Net with CCA	.578	.531	.560	.649
VGG16-ImageNet [29]	.600	—	—	—
VGG16-ImageNet + L^3 -Audio CCA	.493	.458	.464	.618
AVE-Net	.604	.561	.587	.665

Table 1. Cross-modal and intra-modal retrieval. Comparison of our method with unsupervised and supervised baselines in terms of the average nDCG@30 on the AudioSet-Instruments test set. The columns headers denote the modalities of the query and the database, respectively, where *im* stands for *image* and *aud* for *audio*. Our AVE-Net beats all baselines convincingly.

Retrieval performance



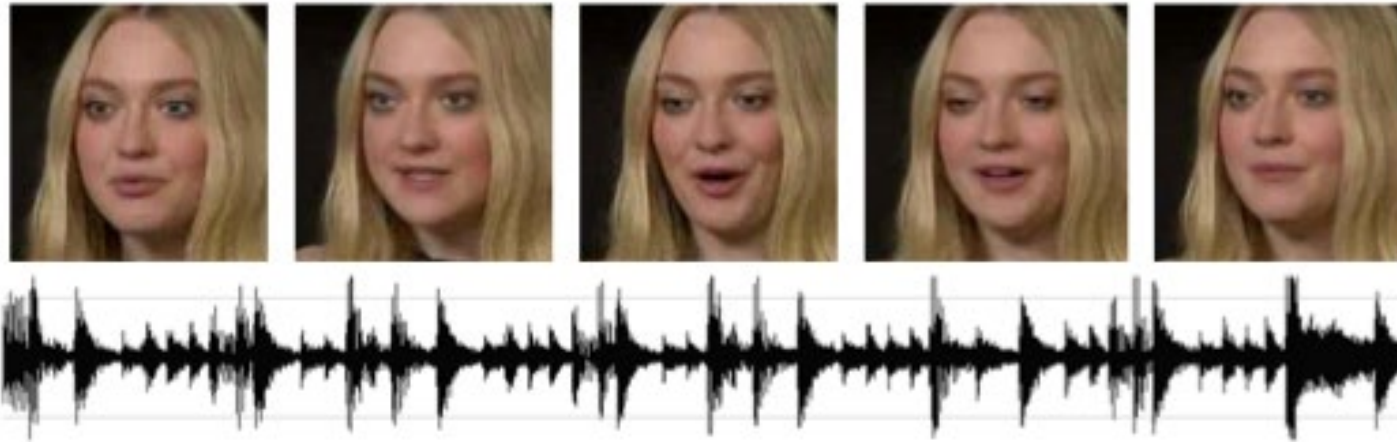
Sound Localization



Faces and Speech

Specialize to talking heads ...

Objective: use faces and voice to learn from each other



- Two types of proxy task:
 1. Predict audio-visual **correspondence**
 2. Predict audio-visual **synchronization**

Faces and Speech

The network can be used for:

- Audio-to-video synchronisation
- Active speaker detection
- Voice-over rejection
- Visual features for lip reading

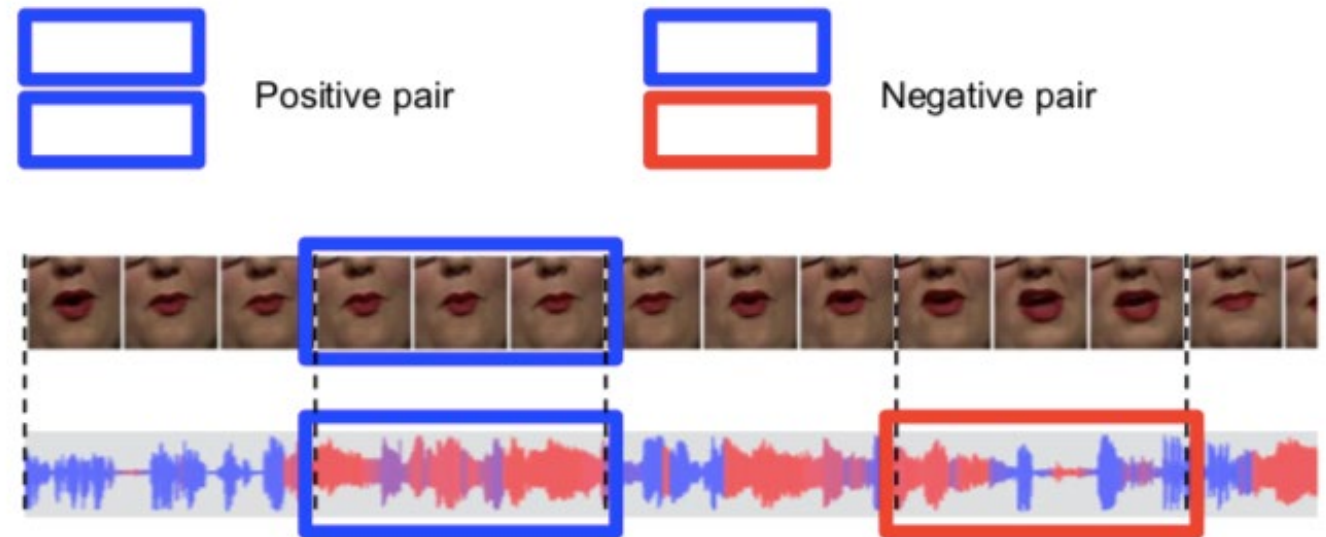


Fig. 5. The process of obtaining *genuine* and *false* audio-video pairs.

J. S. Chung, A. Zisserman, "Out of time: automated lip sync in the wild", Workshop on Multi-view Lip-reading, ACCV 2016

Audio-Visual Scene Analysis

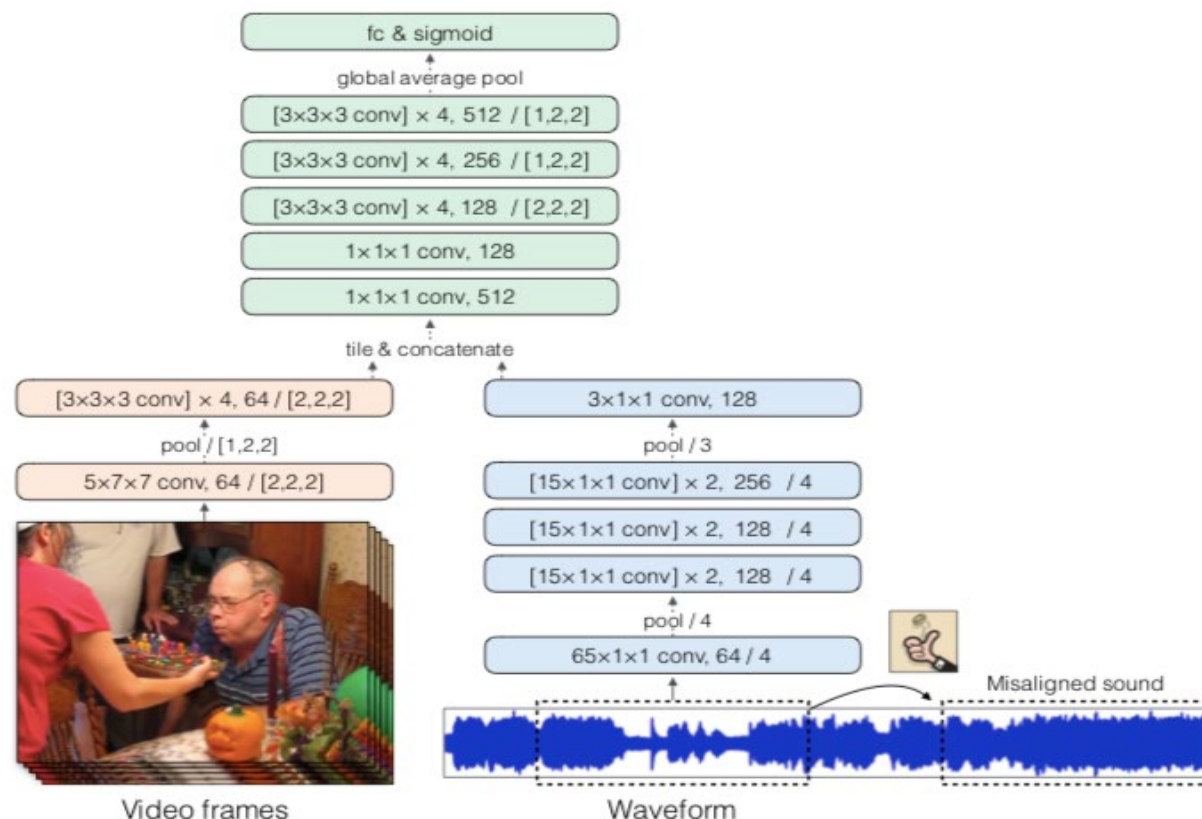


We use this learned representation for:

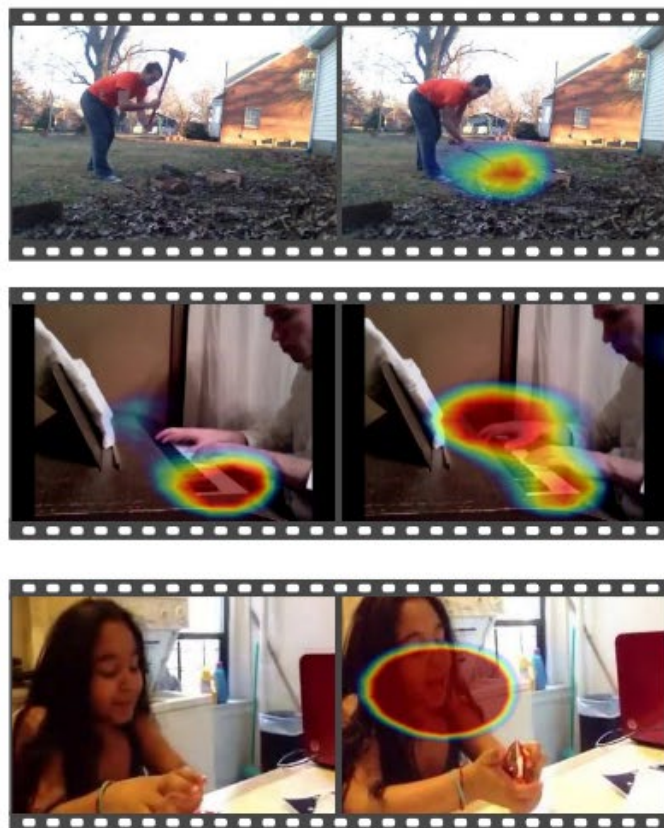
- (a) sound source localization, i.e. visualizing the source of sound in a video;
- (b) audio-visual action recognition;
- (c) on/off-screen audio source separation.

A. Owens & A. Efros, “Audio-Visual Scene Analysis with Self-Supervised Multisensory Features”, ECCV 2018

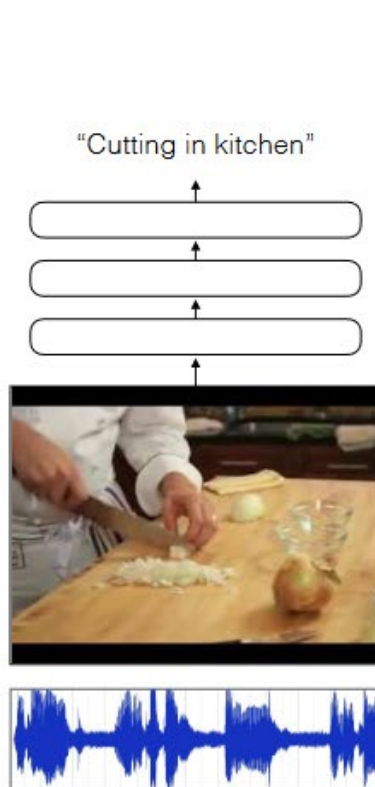
- The visual and audio components of a video signal should be modeled jointly using a fused multisensory representation.
- Such a representation is learned in a self-supervised way, by training a neural network to predict whether video frames and audio are temporally aligned.



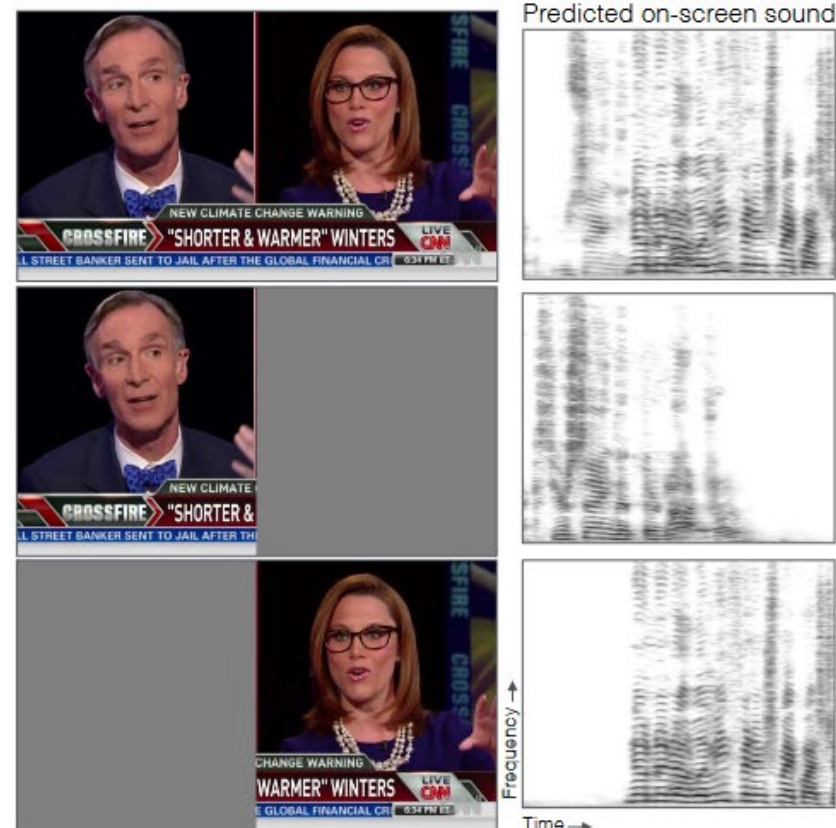
Audio-Visual Scene Analysis



(a) Sound localization

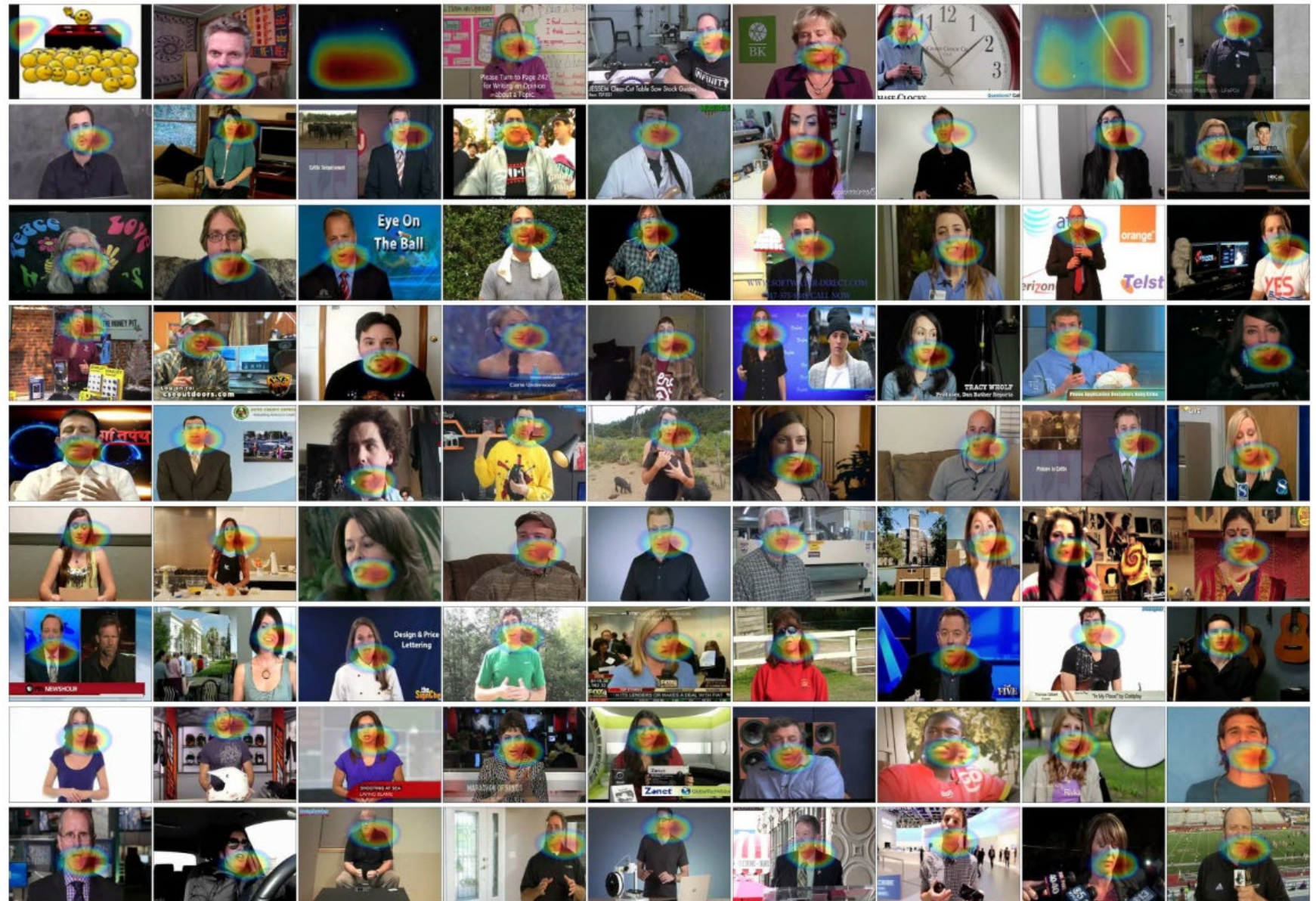


(b) Action recognition



(c) On/off-screen audio separation

Audio-Visual Scene Analysis



A. Owens & A. Efros, "Audio-Visual Scene Analysis with Self-Supervised Multisensory Features", ECCV 2018

Biometric embeddings

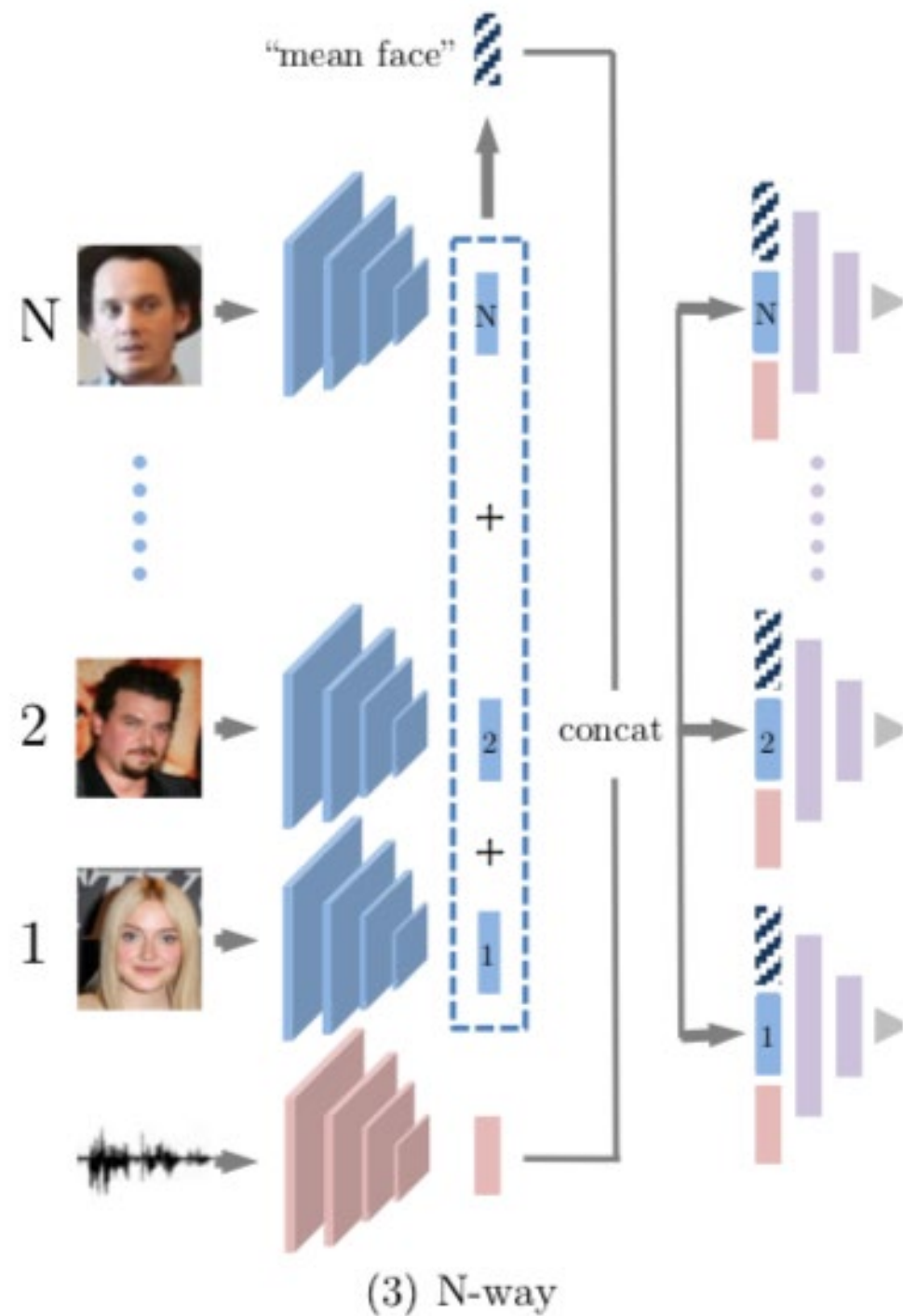
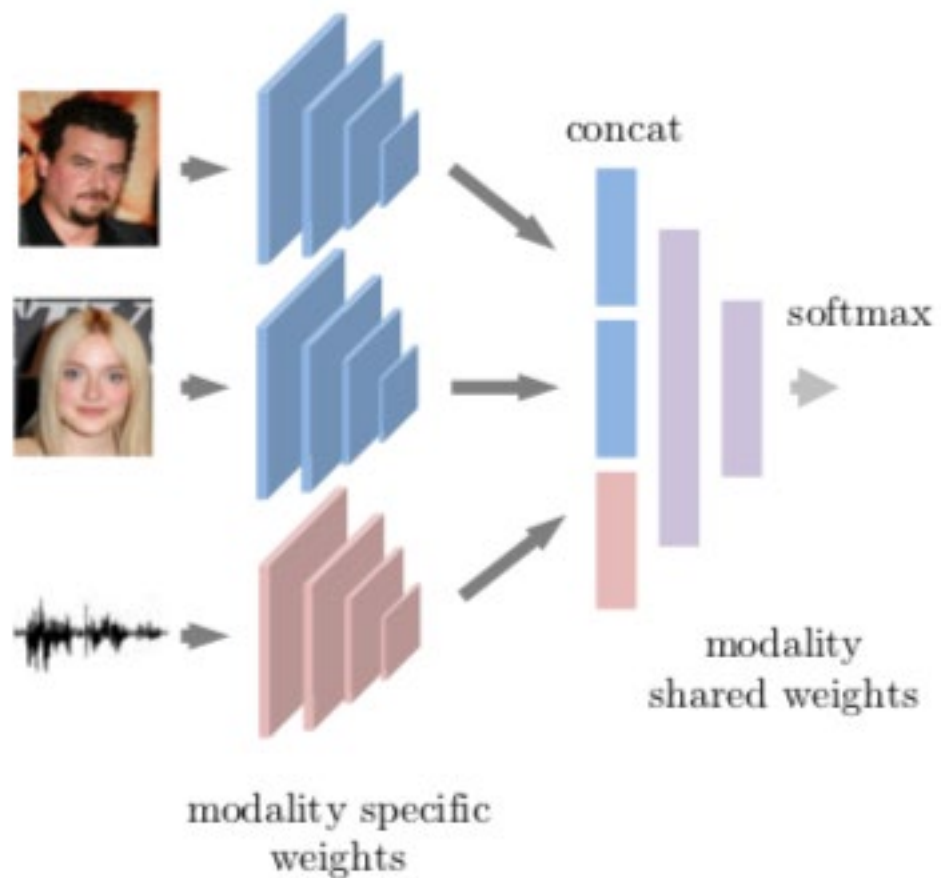
Face/voice correspondence

How much can we infer from the voice about the face and vice versa?

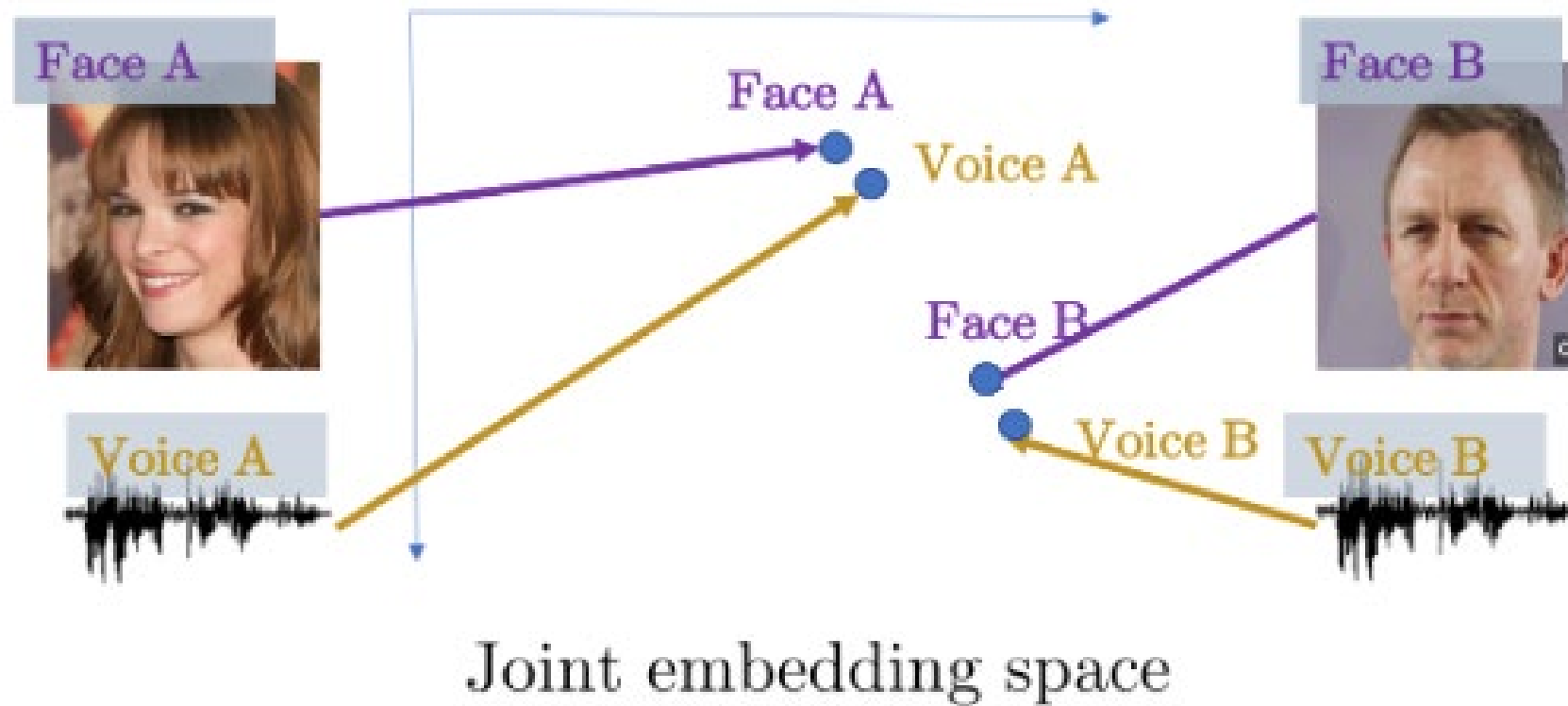
For example in the film “Die Hard”, where the Bruce Willis character (John McLane) emerges from the building towards the end of the film and is instantly able to recognise the cop (Sgt. Al Powell) who he has only spoken to by radio throughout the film, but never seen.

A. Nagrani, S. Albanie, A. Zisserman, “Seeing Voices and Hearing Faces: Cross-modal biometric matching”, CVPR 2018

Biometric embeddings

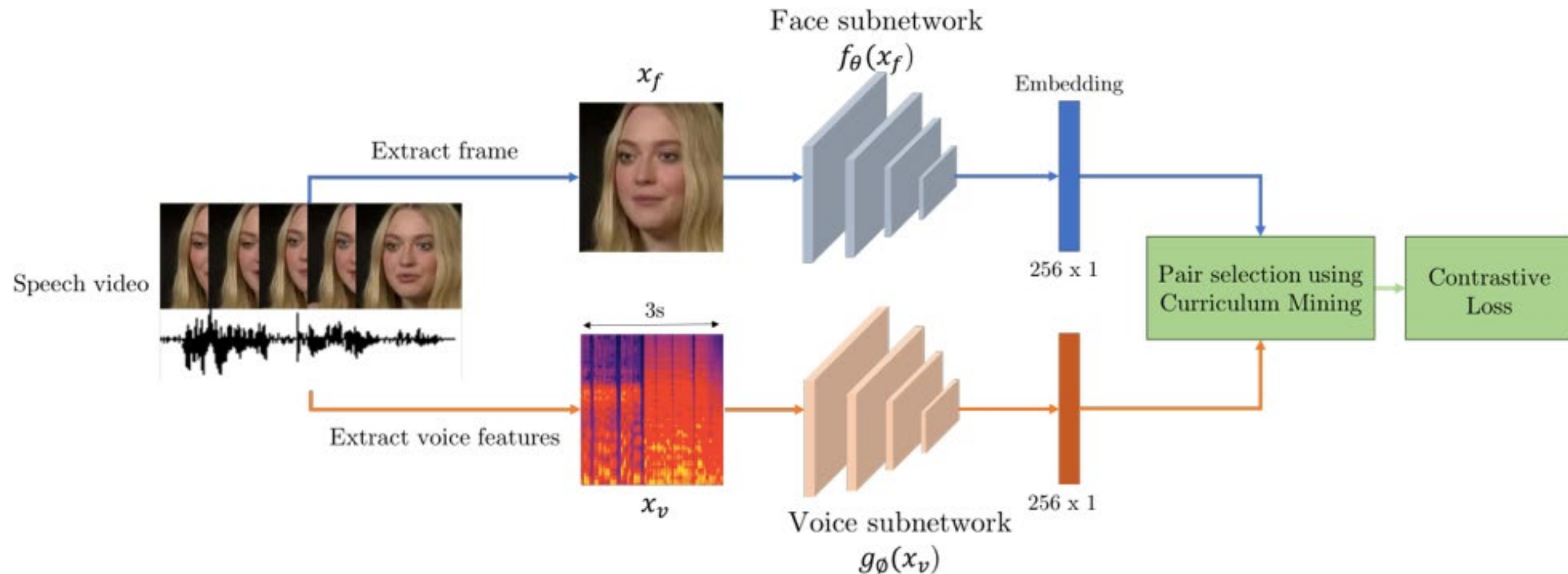


Biometric embeddings



A. Nagrani, S. Albanie, A. Zisserman, "Learnable PINS: Cross-Modal Embeddings for Person Identity", ECCV 2018

Biometric embeddings



Biometric embeddings

Evaluating:

- Cross-modal verification - given a face input and a speech segment, the goal is to determine if they belong to the same identity
- Cross-modal retrieval - Given a single query from one modality, the goal is to retrieve all semantically matching templates from another modality
- One-Shot Learning for TV Show Character Retrieval

To recap ...

- Self-supervised learning from images/video
 - Enable learning without explicit supervision
 - Learns from visual representations – about on par with ImageNet training
- Self-supervised learning from videos with sound
 - Intra- and cross-modal retrieval
 - Learn to localize sounds
 - Tasks are not just a proxy, e.g., synchronization, attention, applicable directly
- Applicable to other domains with paired signals, e.g.
 - Face and voice
 - Infrared/visible
 - RGB/D

Performance

Which pretext task is better?

Method	Pretext Tasks	ImageNet					Places				
		conv1	conv2	conv3	conv4	conv5	conv1	conv2	conv3	conv4	conv5
Places labels [8]	—	—	—	—	—	—	22.1	35.1	40.2	43.3	44.6
ImageNet labels [8]	—	19.3	36.3	44.2	48.3	50.5	22.7	34.8	38.4	39.4	38.7
⇒ Random(Scratch) [8]	—	11.6	17.1	16.9	16.3	14.1	15.7	20.3	19.8	19.1	17.5
ColorfulColorization [18]	Generation	12.5	24.5	30.4	31.5	30.3	16.0	25.7	29.6	30.3	29.7
BiGAN [122]	Generation	17.7	24.5	31.0	29.9	28.0	21.4	26.2	27.1	26.1	24.0
SplitBrain [42]	Generation	17.7	29.3	35.4	35.2	32.8	21.3	30.7	34.0	34.1	32.5
ContextEncoder [19]	Context	14.1	20.7	21.0	19.8	15.5	18.2	23.2	23.4	21.9	18.4
ContextPrediction [41]	Context	16.2	23.3	30.2	31.7	29.6	19.7	26.7	31.9	32.7	30.9
Jigsaw [20]	Context	18.2	28.8	34.0	33.9	27.1	23.0	32.1	35.5	34.8	31.3
Learning2Count [130]	Context	18.0	30.6	34.3	32.5	25.7	23.3	33.9	36.3	34.7	29.6
DeepClustering [44]	Context	13.4	32.3	41.0	39.6	38.2	19.6	33.2	39.2	39.8	34.7

Linear classification on ImageNet and Places datasets on AlexNet feature maps.

Taskonomy

Which pretext task is better for which downstream task?

Method	Pretext Tasks	Classification	Detection	Segmentation
ImageNet Labels [8]	—	79.9	56.8	48.0
Random(Scratch) [8]	—	57.0	44.5	30.1
ContextEncoder [19]	Generation	56.5	44.5	29.7
BiGAN [122]	Generation	60.1	46.9	35.2
ColorfulColorization [18]	Generation	65.9	46.9	35.6
SplitBrain [42]	Generation	67.1	46.7	36.0
RankVideo [38]	Context	63.1	47.2	35.4 [†]
PredictNoise [46]	Context	65.3	49.4	37.1 [†]
JigsawPuzzle [20]	Context	67.6	53.2	37.6
ContextPrediction [41]	Context	65.3	51.1	—
Learning2Count [130]	Context	67.7	51.4	36.6
DeepClustering [44]	Context	73.7	55.4	45.1
WatchingVideo [81]	Free Semantic Label	61.0	52.2	—
CrossDomain [30]	Free Semantic Label	68.0	52.6	—
AmbientSound [154]	Cross Modal	61.3	—	—
TiedToEgoMotion [95]	Cross Modal	—	41.7	—
EgoMotion [94]	Cross Modal	54.2	43.9	—

Classification, detection, and segmentation on PASCAL VOC dataset

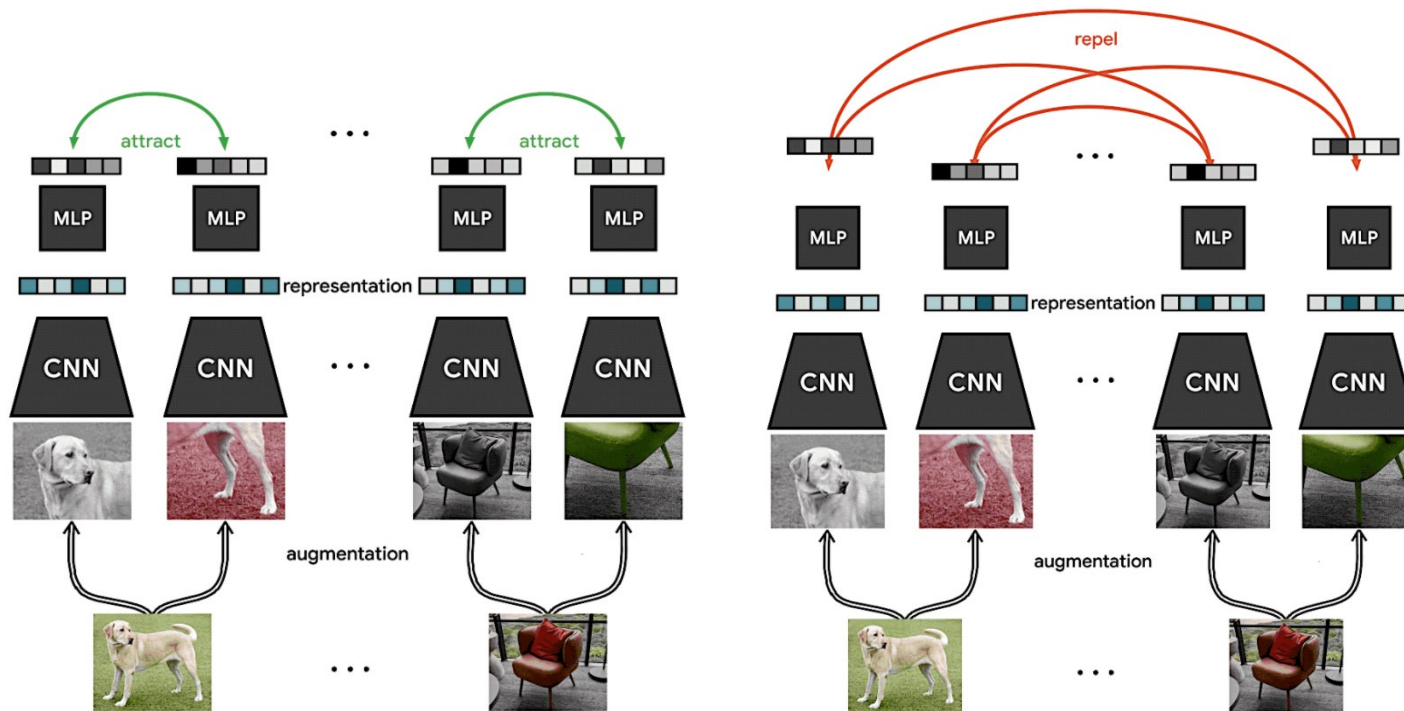
Sax Zamir, Shen, Guibas, Malik, & Savarese. “Taskonomy: Disentangling Task Transfer Learning.” CVPR 2018

Self-supervised, contrastive methods

SimCLR, MoCo, Byol, Supervised Contrastive, SimCLR+

SimCLR – Contrastive Learning

SimCLR learns representations by **maximizing agreement between differently augmented views** of the same data example via a **contrastive loss** in the latent space.



A mini-batch of N examples is randomly sampled and the contrastive prediction task on pairs of augmented examples derived from the mini-batch is defined, resulting in $2N$ data points. Given a positive pair, the other $2(N-1)$ augmented examples within a mini-batch are treated as negative examples.

Then the loss function for a positive pair of examples (i,j) is defined as:

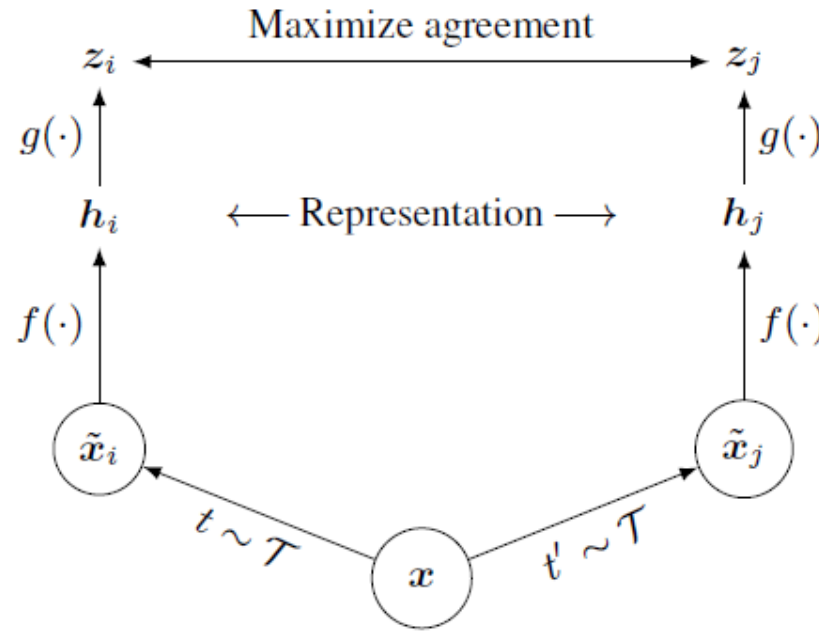
$$\ell_{i,j} = -\log \frac{\exp(\text{sim}(z_i, z_j)/\tau)}{\sum_{k=1}^{2N} \mathbb{1}_{[k \neq i]} \exp(\text{sim}(z_i, z_k)/\tau)}$$

The final loss is computed across all positive pairs, both (i,j) and (j,i) , in a mini-batch.

The training requires a big batch size.

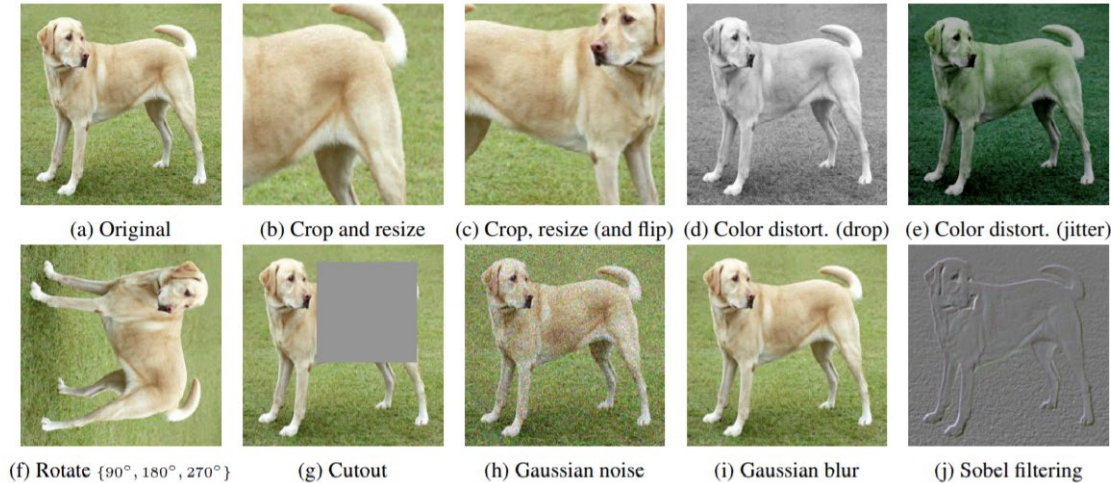
Ting Chen et al. "A simple framework for contrastive learning of visual representations." *ICML* 2020.

SimCLR



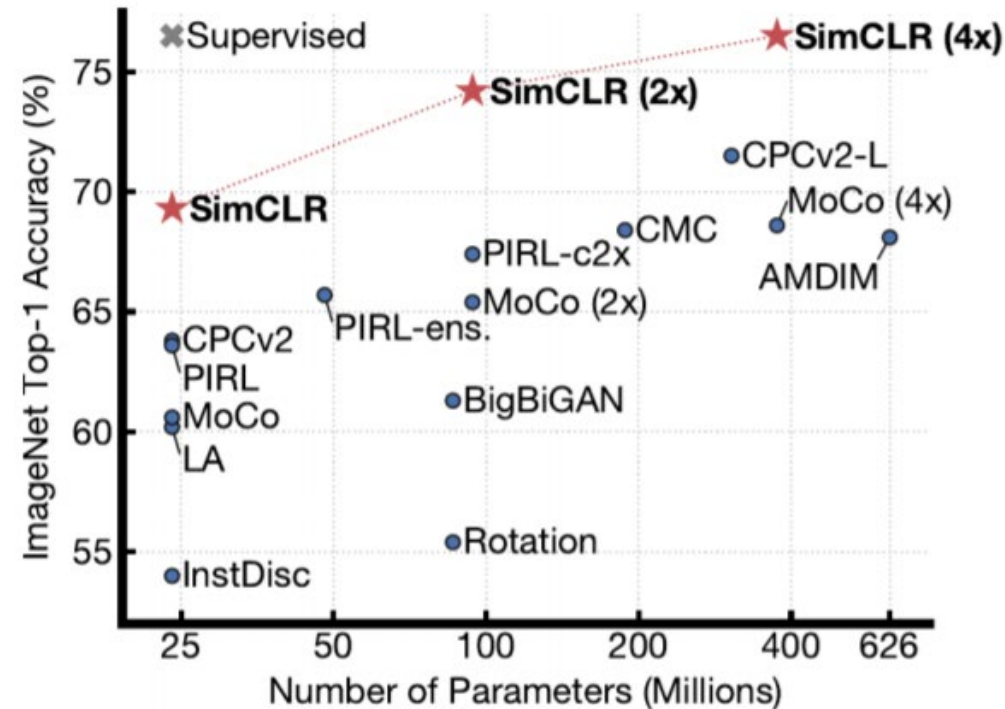
- **Two separate data augmentation** operators are **sampled from the same family of augmentations** ($t \sim \mathcal{T}$ and $t' \sim \mathcal{T}$) and applied to each data example to obtain **two correlated views**.
- A base encoder network $f(\cdot)$ and a projection head $g(\cdot)$ are trained to maximize agreement using a contrastive loss.
- After training is completed, we throw away the projection head $g(\cdot)$ and use encoder $f(\cdot)$ and representation h for downstream tasks.
- In fact the paper shows **h is a more discriminant feature** than z for downstream tasks.

SimCLR, Data Augmentations



- **Figure (left):** Illustrations of the studied data augmentation operators. Each augmentation can transform data stochastically with some internal parameters (e.g. rotation degree, noise level).
- **Figure (right):** Linear evaluation (ImageNet top-1 accuracy) under **individual or composition of data augmentations**, applied only to one branch. For all columns but the last, diagonal entries correspond to single transformation, and off-diagonals correspond to composition of two transformations (applied sequentially).
- The last column reflects the average over the row: we observe that no single transformation suffices to learn good representations, even though the model can almost perfectly identify the positive pairs in the contrastive task.
- When composing augmentations, the contrastive prediction task becomes harder, but the quality of representation improves dramatically.

SimCLR results (1)



- **ImageNet Top-1 accuracy of linear classifiers** trained on representations learned with different self-supervised methods (pretrained on ImageNet).
- Gray cross indicates supervised ResNet-50.
- SimCLR, is shown in bold. ResNet-50 architecture is used in 3 different hidden layer widths (width multipliers of 1x, 2x, and 4x)

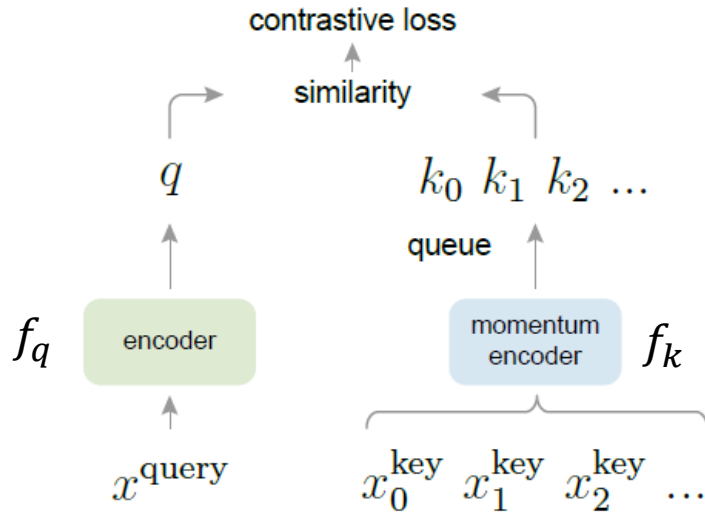
SimCLR results (2)

*The linear evaluation protocol refers to a linear classifier trained on top of the frozen base network, and test accuracy is used as a proxy for representation quality.

	Food	CIFAR10	CIFAR100	Birdsnap	SUN397	Cars	Aircraft	VOC2007	DTD	Pets	Caltech-101	Flowers
<i>Linear evaluation:</i>												
SimCLR (ours)	76.9	95.3	80.2	48.4	65.9	60.0	61.2	84.2	78.9	89.2	93.9	95.0
Supervised	75.2	95.7	81.2	56.4	64.9	68.8	63.8	83.8	78.7	92.3	94.1	94.2
<i>Fine-tuned:</i>												
SimCLR (ours)	89.4	98.6	89.0	78.2	68.1	92.1	87.0	86.6	77.8	92.1	94.1	97.6
Supervised	88.7	98.3	88.7	77.8	67.0	91.4	88.0	86.5	78.8	93.2	94.2	98.0
Random init	88.3	96.0	81.9	77.0	53.7	91.3	84.8	69.4	64.1	82.7	72.5	92.5

- Comparison of **transfer learning performance** of our self-supervised approach with supervised baselines across 12 natural image classification datasets, for ResNet-50 (4×) models pretrained on ImageNet.
- Results not significantly worse than the best ($p > 0.05$, permutation test) are shown in bold.
- When fine-tuned, the self-supervised **SimCLR model significantly outperforms the supervised baseline on 5 datasets**, whereas the supervised baseline is superior on only 2 (i.e. Pets and Flowers). On the remaining 5 datasets, the models are statistically tied.

Self-Supervised Learning: MoCo – Momentum Contrast



Momentum Contrast (MoCo) trains a visual representation encoder by **matching an encoded query q to a dictionary of encoded keys using a contrastive loss**. The dictionary keys are defined on-the-fly by a set of data samples.

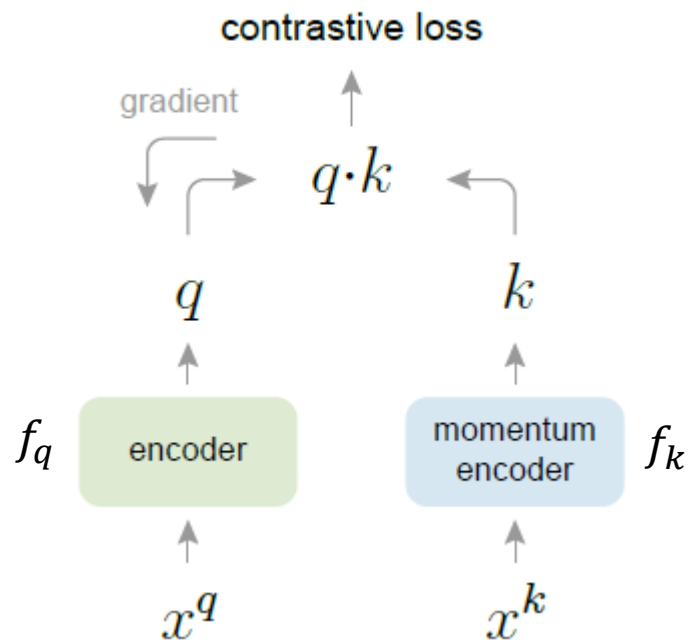
The **dictionary is built as a queue**, with the current mini-batch enqueued and the oldest mini-batch dequeued, decoupling it from the mini-batch size. The keys are encoded by a slowly progressing encoder, driven by a momentum update with the query encoder.

$$\mathcal{L}_q = -\log \frac{\exp(q \cdot k_+ / \tau)}{\sum_{i=0}^K \exp(q \cdot k_i / \tau)}$$

A **contrastive loss** is a function whose **value is low when q is similar to its positive key k_+ and dissimilar to all other keys considered negative keys for q** . Intuitively, this loss is the log loss of a $(K+1)$ -way softmax-based classifier that **tries to classify q as k_+** .

Kaiming He et al. "Momentum contrast for unsupervised visual representation learning." *CVPR* 2020.

MoCo



Using a **queue** can make the **dictionary large**, but it also makes it **intractable to update the key encoder by back-propagation** (the gradient should propagate to all samples in the queue). Formally, denoting the parameters of the key encoder f_k as θ_k and those of the query encoder f_q as θ_q , f_k is updated by:

$$\theta_k \leftarrow m\theta_k + (1 - m)\theta_q$$

where $m \in [0.1)$ is the momentum coefficient.

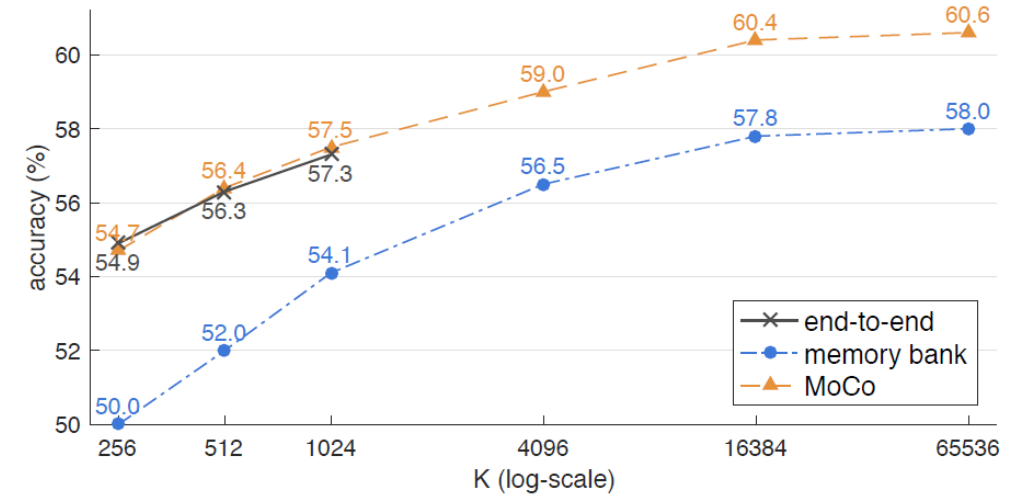
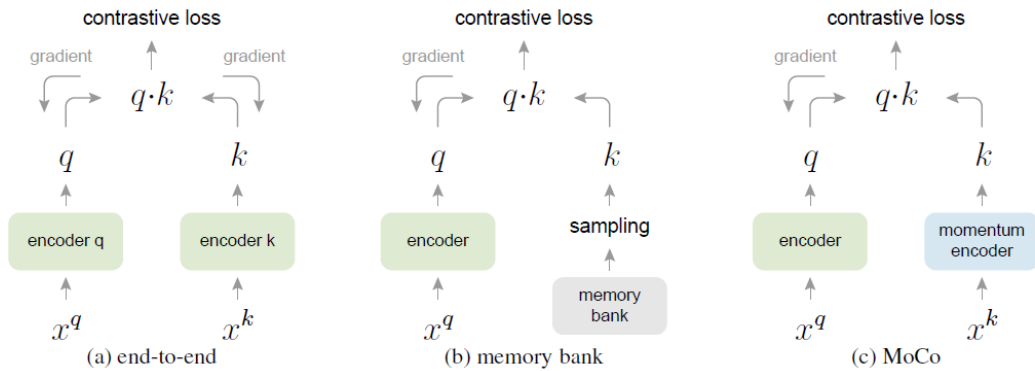
Only the parameters θ_q are updated by back-propagation.

The momentum update in eqn. above makes f_k evolve more smoothly than f_q .

As a result, though the keys in the queue are encoded by different encoders (in different mini-batches), the difference among these encoders can be made small.

The main hypothesis is that **good features can be learned by a large dictionary** that covers a **rich set of negative samples**, while the encoder for the dictionary keys is kept as consistent as possible despite its evolution.

MoCo Results (1)

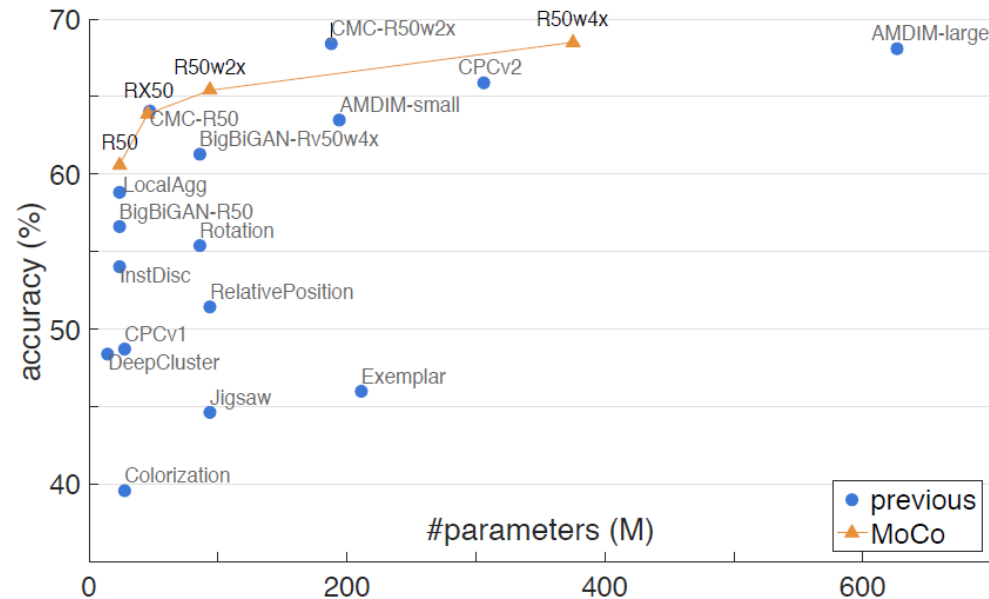


On the figure at left we illustrate one pair of query and key. The three illustrated mechanisms differ in **how the keys are maintained and how the key encoder is updated**.

(a): The encoders for computing the query and key representations are updated **end-to-end by back-propagation** (the two encoders can be different). (b): The key representations are sampled from a **memory bank**. (c): MoCo encodes the new **keys on-the-fly by a momentum-updated encoder**, and maintains a **queue** (not illustrated in this figure) of keys.

- In the figure at right the comparison of three contrastive loss mechanisms under for ImageNet classification task.
- The **end-to-end mechanism** performs similarly to MoCo when K is small. However, the dictionary size is limited by the mini-batch size due to the end-to-end requirement.
- The **memory bank** mechanism can support a larger dictionary size. But it is 2.6% worse than **MoCo**. This is in line with this hypothesis: the keys in the memory bank are from very different encoders all over the past epoch and they are not consistent.

MoCo Results (2)



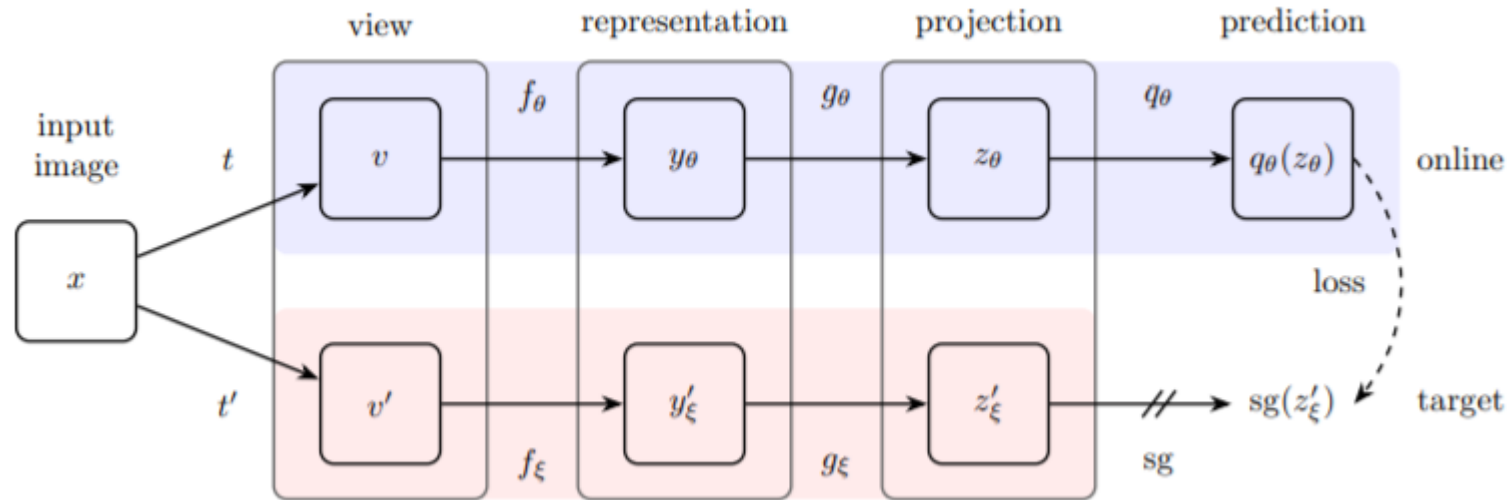
method	architecture	#params (M)	accuracy (%)
Exemplar [17]	R50w3x	211	46.0 [38]
RelativePosition [13]	R50w2x	94	51.4 [38]
Jigsaw [45]	R50w2x	94	44.6 [38]
Rotation [19]	Rv50w4x	86	55.4 [38]
Colorization [64]	R101*	28	39.6 [14]
DeepCluster [3]	VGG [53]	15	48.4 [4]
BigBiGAN [16]	R50	24	56.6
	Rv50w4x	86	61.3

methods based on contrastive learning follow:

InstDisc [61]	R50	24	54.0
LocalAgg [66]	R50	24	58.8
CPC v1 [46]	R101*	28	48.7
CPC v2 [35]	R170* _{wider}	303	65.9
CMC [56]	R50 _{L+ab}	47	64.1 [†]
	R50w2x _{L+ab}	188	68.4 [†]
AMDIM [2]	AMDIM _{small}	194	63.5 [†]
	AMDIM _{large}	626	68.1 [†]
MoCo	R50	24	60.6
	RX50	46	63.9
	R50w2x	94	65.4
	R50w4x	375	68.6

- Comparison under the **linear classification protocol on ImageNet** (the figure visualizes the table): all are reported as **unsupervised pre-training on the ImageNet-1M training set**, followed by supervised linear classification trained on frozen features, evaluated on the validation set. The parameter counts are those of the feature extractors.
- MoCo can outperform its supervised pre-training counterpart in 7 detection/segmentation tasks on PASCAL VOC, COCO, and other datasets, sometimes surpassing it by large margins.
- This suggests that the **gap between unsupervised and supervised representation learning has been largely closed** in many vision tasks.

BYOL Bootstrap Your Own Latent



- BYOL is a method that revisits advancements proposed in previous methods (MoCo and SimCLR) for positive learning.
- Similarly to SimCLR, it makes use of **strong data augmentation** transformations on the image x and the **feature representation used at inference time is f** (all the other blocks are discarded). Whereas, similarly to MoCo it makes use of an **online network t** and a **target (exponential) mean network t'** .
- Moreover, BYOL **is not a contrastive approach**, it **maximizes the similarity** metric between the target and online predictions by means of mean squared error.

BYOL

Process overview:

We take one unlabelled image and generate two views, which should share the same label.

Then, Byol uses the top-view image, pass it into a standard network-predictor. Instead of adopting ground truth labels we pass the second image through another network that we called label generator network which is a delayed replica of the upper one except removing the predictor module.

Then, we use the latent output of the bottom part as the labels for the upper part, in practice we compute the mean squared error between the 2 latents.

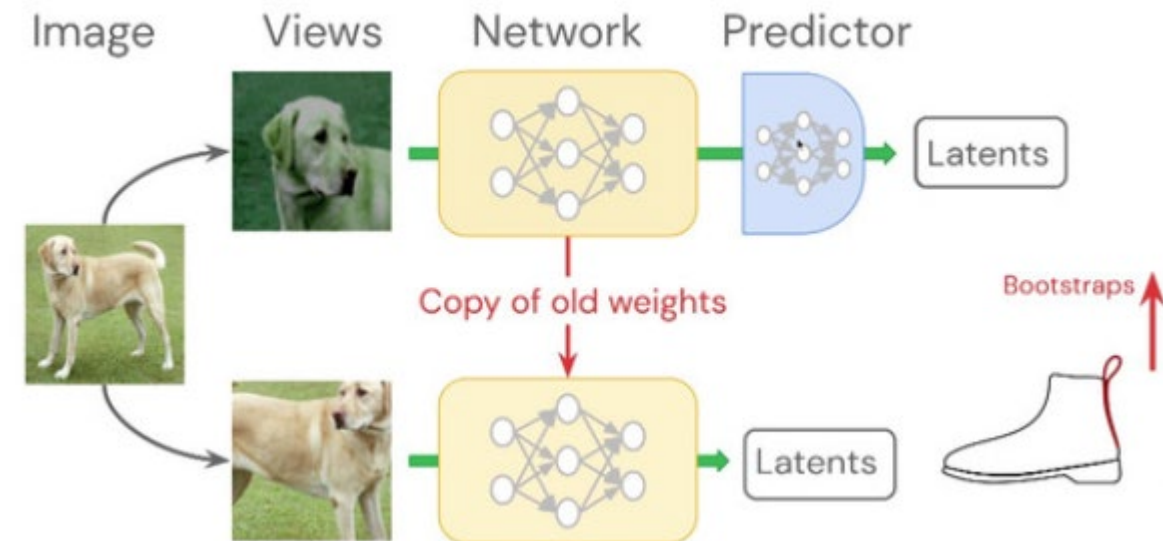
We train then the upper part in supervised way.

Key ingredients:

- Image transformations
- Target network
- Additional predictor on online network (if we do not use it, the representation collapses)

Takeaways:

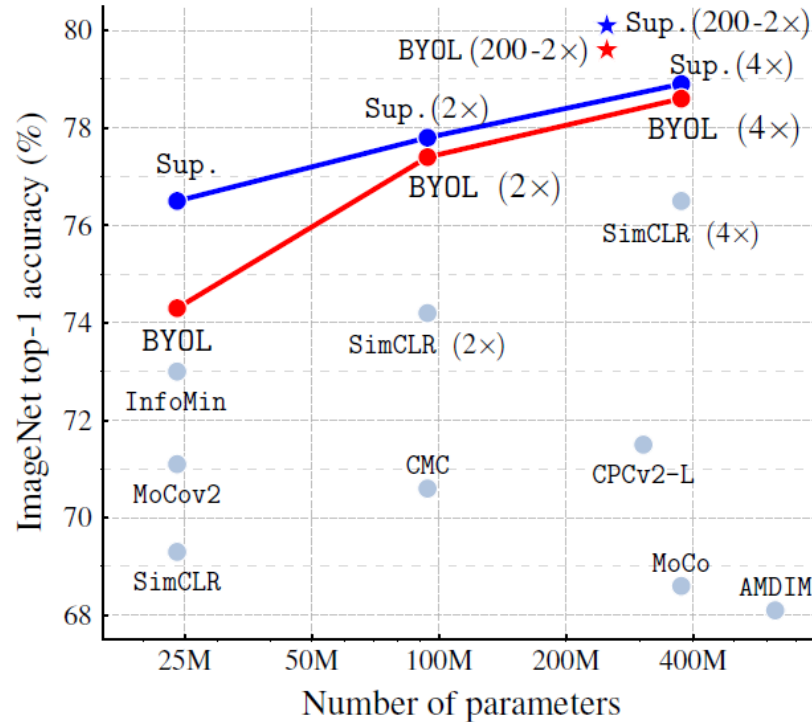
- Does not use **negative examples**
- **More resilient to transformation and batch size** than contrastive methods



Grill et al. "Bootstrap Your Own Latent A New Approach to Self-Supervised Learning." *NeurIPS* 2020.

https://neurips.cc/virtual/2020/public/poster_f3ada80d5c4ee70142b17b8192b2958e.html

BYOL results (1)



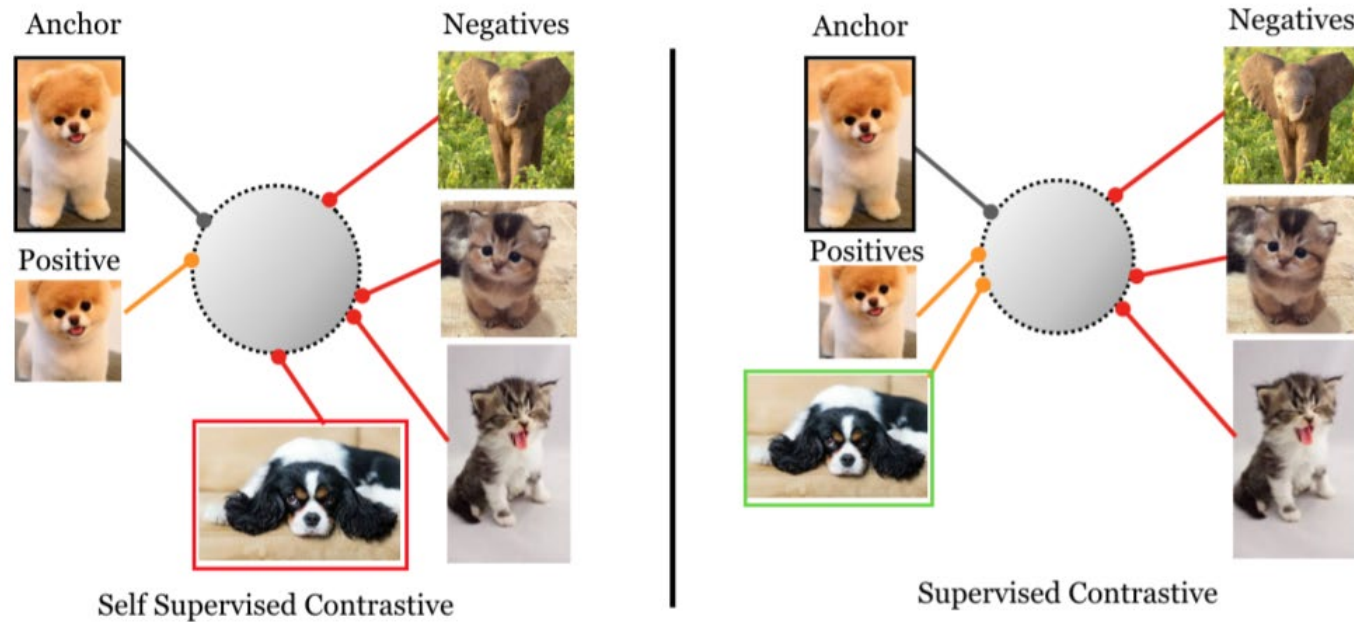
- **Performance of BYOL on ImageNet** (linear evaluation) using ResNet-50 and the best architecture ResNet-200 (4x), compared to other unsupervised and supervised (Sup.) baselines.
- BYOL achieves higher performance than state-of-the-art contrastive methods **without using negative pairs**. It iteratively bootstraps the outputs of a network to serve as targets for an enhanced representation.

BYOL results (2)

Method	Food101	CIFAR10	CIFAR100	Birdsnap	SUN397	Cars	Aircraft	VOC2007	DTD	Pets	Caltech-101	Flowers
<i>Linear evaluation:</i>												
BYOL (ours)	75.3	91.3	78.4	57.2	62.2	67.8	60.6	82.5	75.5	90.4	94.2	96.1
SimCLR (repro)	72.8	90.5	74.4	42.4	60.6	49.3	49.8	81.4	75.7	84.6	89.3	92.6
SimCLR [8]	68.4	90.6	71.6	37.4	58.8	50.3	50.3	80.5	74.5	83.6	90.3	91.2
Supervised-IN [8]	72.3	93.6	78.3	53.7	61.9	66.7	61.0	82.8	74.9	91.5	94.5	94.7
<i>Fine-tuned:</i>												
BYOL (ours)	88.5	97.8	86.1	76.3	63.7	91.6	88.1	85.4	76.2	91.7	93.8	97.0
SimCLR (repro)	87.5	97.4	85.3	75.0	63.9	91.4	87.6	84.5	75.4	89.4	91.7	96.6
SimCLR [8]	88.2	97.7	85.9	75.9	63.5	91.3	88.1	84.1	73.2	89.2	92.1	97.0
Supervised-IN [8]	88.3	97.5	86.4	75.8	64.3	92.1	86.0	85.0	74.6	92.1	93.3	97.6
Random init [8]	86.9	95.9	80.2	76.1	53.6	91.4	85.9	67.3	64.8	81.5	72.6	92.0

- The Table reports results for both linear evaluation and fine-tuning.
- **BYOL outperforms SimCLR** on all benchmarks and the Supervised-IN (features learned on ImageNet(IN)) baseline on 7 out of the 12 benchmarks, providing only slightly worse performance on the 5 remaining benchmarks.
- BYOL's representation can be transferred over to small images, e.g., CIFAR, landscapes, e.g., SUN397 or VOC2007, and textures, e.g., DTD

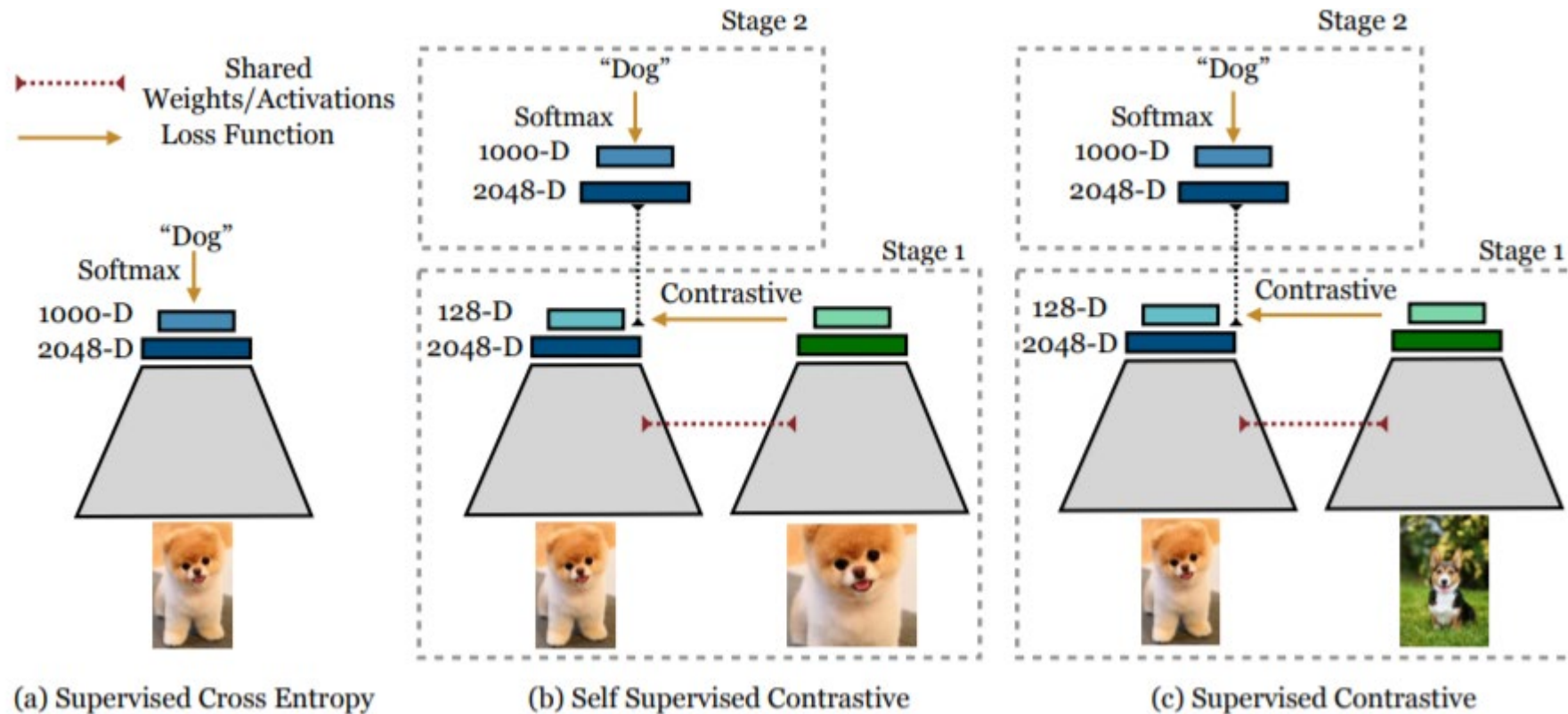
Supervised Contrastive Learning



- **Supervised vs. self-supervised contrastive losses:**
- The *self-supervised contrastive loss* contrasts a single positive for each anchor (i.e., an augmented version of the same image) against a set of negatives consisting of the entire remainder of the batch.
- The *supervised contrastive loss* (right), however, contrasts the set of all samples from the same class as positives against the negatives from the remainder of the batch. As demonstrated by the photo of the black and white puppy, taking class label information into account results in an embedding space where elements of the same class are more closely aligned than in the self-supervised case

Supervised Contrastive Learning

SupCon aims at merging the best of two worlds, supervised learning (a) and recent contrastive based self-supervised learning (b). In fact they propose a supervised contrastive loss (c).



Supervised Contrastive Learning

SupCon aims at merging the best of two worlds, supervised learning and recent contrastive based self-supervised learning. In fact they propose a supervised contrastive loss, the two version are shown here

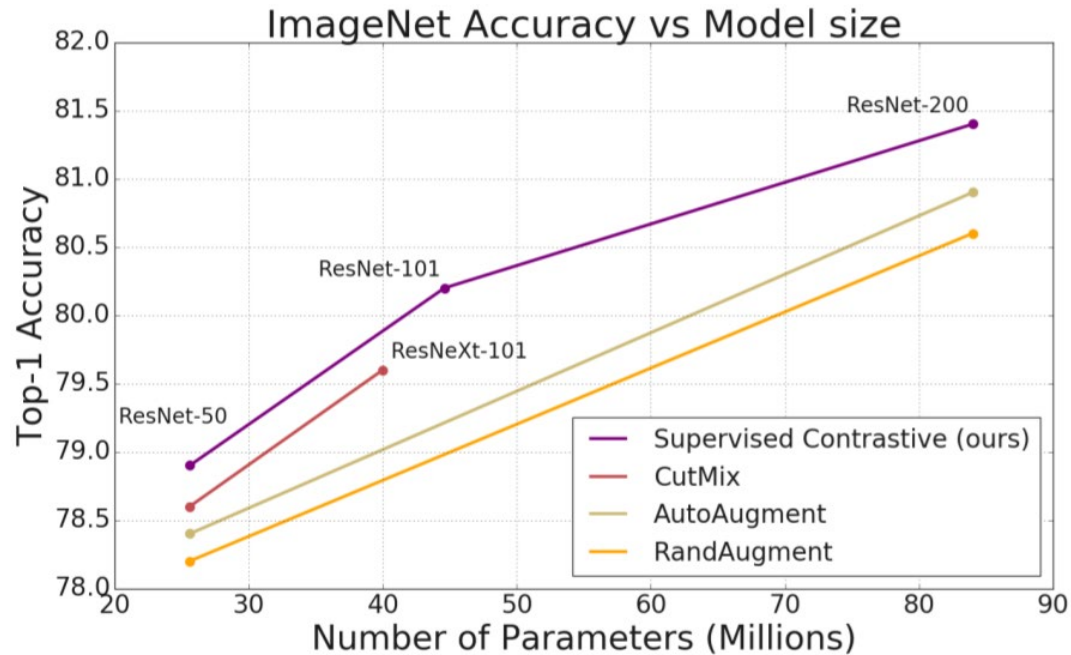
$$\mathcal{L}_{out}^{sup} = \sum_{i \in I} \mathcal{L}_{out,i}^{sup} = \sum_{i \in I} \frac{-1}{|P(i)|} \sum_{p \in P(i)} \log \frac{\exp(z_i \cdot z_p / \tau)}{\sum_{a \in A(i)} \exp(z_i \cdot z_a / \tau)}$$
$$\mathcal{L}_{in}^{sup} = \sum_{i \in I} \mathcal{L}_{in,i}^{sup} = \sum_{i \in I} -\log \left\{ \frac{1}{|P(i)|} \sum_{p \in P(i)} \frac{\exp(z_i \cdot z_p / \tau)}{\sum_{a \in A(i)} \exp(z_i \cdot z_a / \tau)} \right\}$$

There are **3 benefits** from these losses: generalization to an arbitrary number of positives, contrastive power increases with more negatives (big batch size) and intrinsic ability to perform hard positive/negative mining.

Finally, the connection between these losses and the widely known triplet loss and N-pair loss are investigated, which are special cases of the proposed losses. Regarding contrastive loss, it is the degenerate case where each image in the dataset has a different label.

Their experiments show these losses outperform both cross entropy and contrastive loss.

Supervised Contrastive Learning Results



Dataset	SimCLR[3]	Cross-Entropy	Max-Margin [32]	SupCon
CIFAR10	93.6	95.0	92.4	96.0
CIFAR100	70.7	75.3	70.5	76.5
ImageNet	70.2	78.2	78.0	78.7

Table 2: Top-1 classification accuracy on ResNet-50 [17] for various datasets. We compare cross-entropy training, unsupervised representation learning (SimCLR [3]), max-margin classifiers [32] and SupCon (ours). We re-implemented and tuned hyperparameters for all baseline numbers except margin classifiers where we report published results. Note that the CIFAR-10 and CIFAR-100 results are from our PyTorch implementation and ImageNet from our TensorFlow implementation.

Loss	Architecture	Augmentation	Top-1	Top-5
Cross-Entropy (baseline)	ResNet-50	MixUp [61]	77.4	93.6
Cross-Entropy (baseline)	ResNet-50	CutMix [60]	78.6	94.1
Cross-Entropy (baseline)	ResNet-50	AutoAugment [5]	78.2	92.9
Cross-Entropy (our impl.)	ResNet-50	AutoAugment [30]	77.6	95.3
SupCon	ResNet-50	AutoAugment [5]	78.7	94.3
Cross-Entropy (baseline)	ResNet-200	AutoAugment [5]	80.6	95.3
Cross-Entropy (our impl.)	ResNet-200	Stacked RandAugment [49]	80.9	95.2
SupCon	ResNet-200	Stacked RandAugment [49]	81.4	95.9
SupCon	ResNet-101	Stacked RandAugment [49]	80.2	94.7

Table 3: Top-1/Top-5 accuracy results on ImageNet for AutoAugment [5] with ResNet-50 and for Stacked RandAugment [49] with ResNet-101 and ResNet-200. The baseline numbers are taken from the referenced papers, and we also re-implement cross-entropy.

- SupCon loss consistently outperforms *cross-entropy with standard data augmentations*. **Figure on the left** shows the top-1 accuracy for the ImageNet dataset, on ResNet-50, ResNet-101 and ResNet200, and compare against AutoAugment, RandAugment and CutMix.
- Table 2** shows that SupCon generalizes better than cross-entropy, margin classifiers (with use of labels) and unsupervised contrastive learning techniques on CIFAR-10, CIFAR-100 and ImageNet datasets.
- Table 3** shows results for ResNet-50 and ResNet-200 for ImageNet: SupCon achieves a new state of the art accuracy of 78.7% on ResNet-50 with AutoAugment

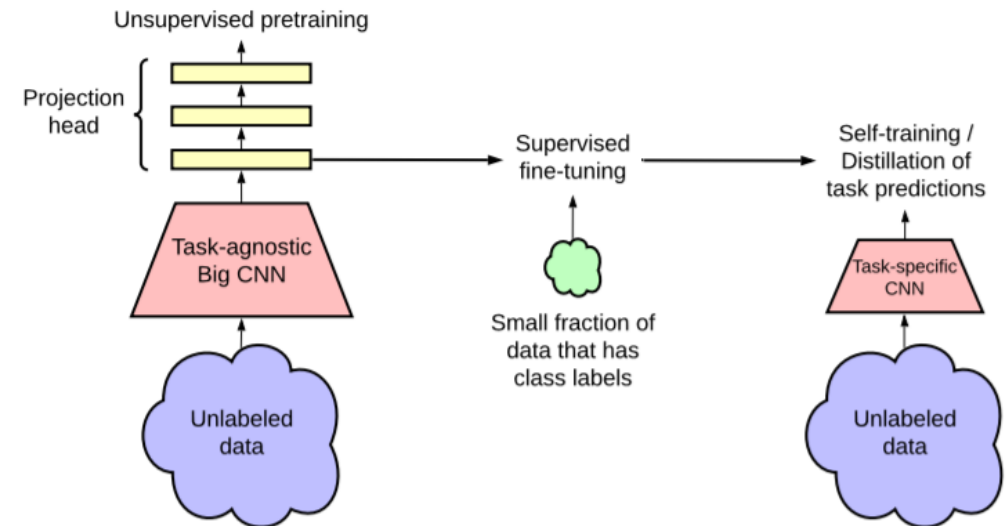
SimCLR+

This work is a first attempt to exploit current success in contrastive self-supervised learning for *semi-supervised learning*. The work presents a simple method for semi-supervised ImageNet Classification in 3 steps: 1) Unsupervised pre-training, 2) Supervised fine-tuning, and 3) Distillation with unlabeled data.

Starting from the well-established SimCLR framework it's shown that feature representations obtained with self-supervision can effectively enhance the semi-supervised learning process where only a limited part of the dataset is annotated (typically 1% to 10% labelled images).

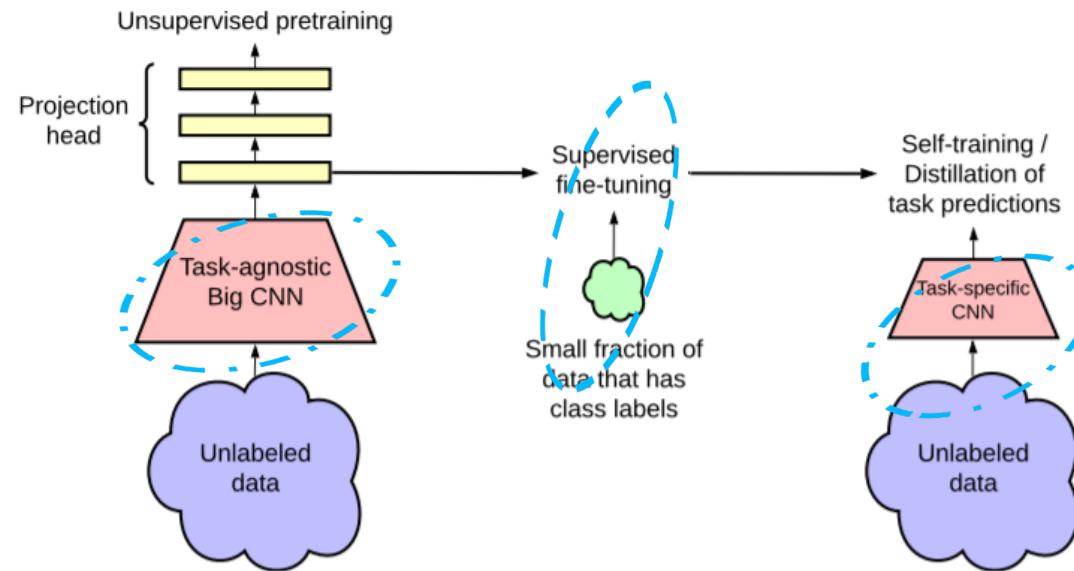
Specifically:

- Bigger self-supervised models are more label efficient, performing significantly better when fine-tuned on only a few labeled examples.
- Distillation of knowledge to smaller task-specific models is an effective way to avoid using big models.
- Deeper projection head (recall the $g(.)$ block in SimCLR) not only improves the representation quality measured by linear evaluation, but also improves semi-supervised performance when fine-tuning.



Chen et al. "Big Self-Supervised Models are Strong Semi-Supervised Learners." *NeurIPS* 2020.

SimCLR+



- The proposed semi-supervised learning framework leverages unlabeled data in both **task-agnostic** and **task-specific** ways.
- **First**, the unlabeled data is used, it is in a *task-agnostic* way, for learning general (visual) representations via unsupervised pretraining.
- The general representations are then adapted for a specific task via *supervised fine-tuning*.
- **Second**, the unlabeled data is used, it is in a *task-specific* way, for further improving predictive performance and obtaining a compact model: to this end, we train student networks on the unlabeled data with imputed labels from the fine-tuned teacher network.
- This method can be summarized in three main steps: **pretrain, fine-tune, and then distill**.

Chen et al. "Big Self-Supervised Models are Strong Semi-Supervised Learners." *NeurIPS* 2020.

Task-specific distillation loss

- Unlabeled case

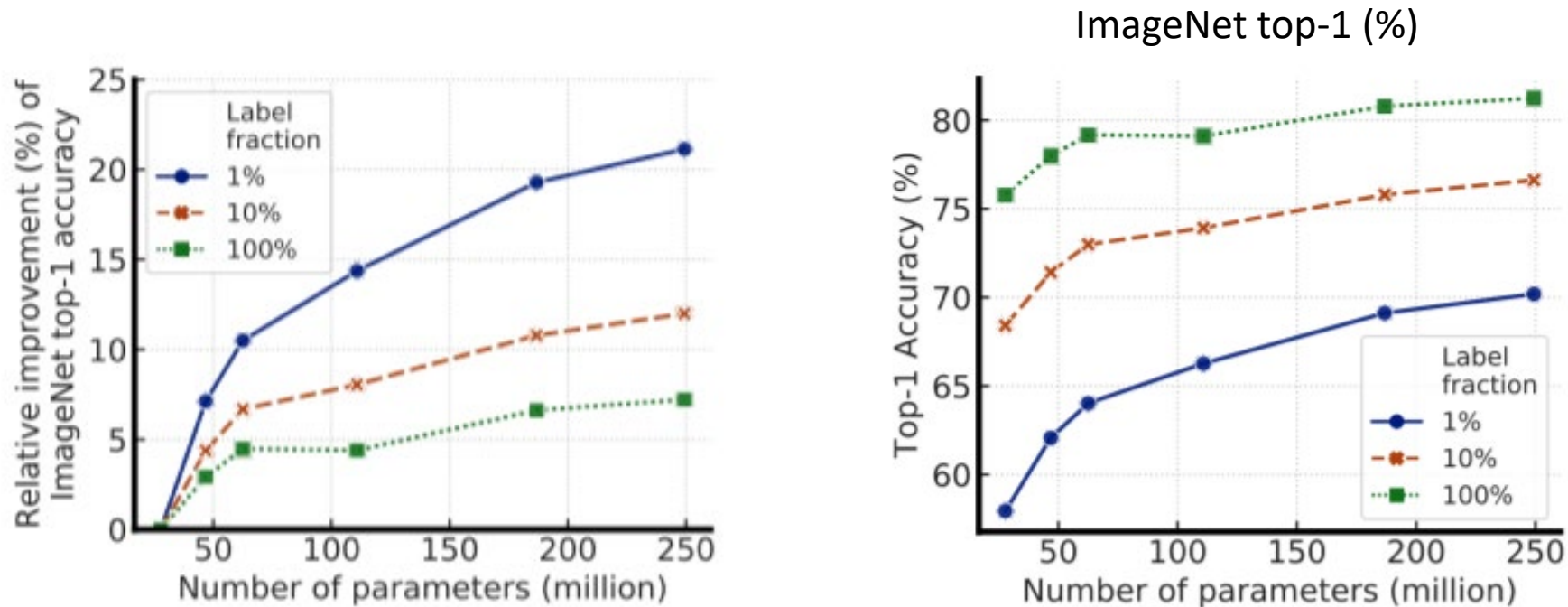
$$\mathcal{L}^{\text{distill}} = - \sum_{\mathbf{x}_i \in \mathcal{D}} \left[\sum_y P^T(y|\mathbf{x}_i; \tau) \log P^S(y|\mathbf{x}_i; \tau) \right]$$

- Semi-supervised case (fine tuning)

$$\mathcal{L} = -(1 - \alpha) \sum_{(\mathbf{x}_i, y_i) \in \mathcal{D}^L} \left[\log P^S(y_i|\mathbf{x}_i) \right] - \alpha \sum_{\mathbf{x}_i \in \mathcal{D}} \left[\sum_y P^T(y|\mathbf{x}_i; \tau) \log P^S(y|\mathbf{x}_i; \tau) \right]$$

SimCLR+ - Bigger models are more label-efficient

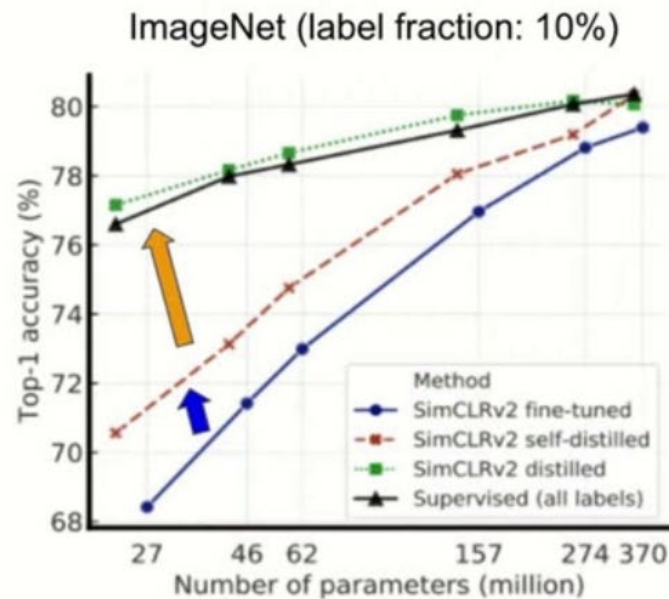
- Using pre-training + fine-tuning, “the fewer the labels, the bigger the model”: bigger models yield larger gains when fine-tuning with fewer labeled examples.
- Increasing the size of model by 10x, it reduces required labels to achieve certain accuracy by 10x



Chen et al. "Big Self-Supervised Models are Strong Semi-Supervised Learners." *NeurIPS* 2020.

SimCLR+: Distillation with unlabeled data improves all model size

- Using pre-training + fine-tuning, “the fewer the labels, the bigger the model”: bigger models yield larger gains when fine-tuning with fewer labeled examples.
- Increasing the size of model by 10x, it reduces required labels to achieve certain accuracy by 10x
- Unlabeled data are crucial for distillation.



Self-distillation / self-training

Distillation from the biggest self-distilled model

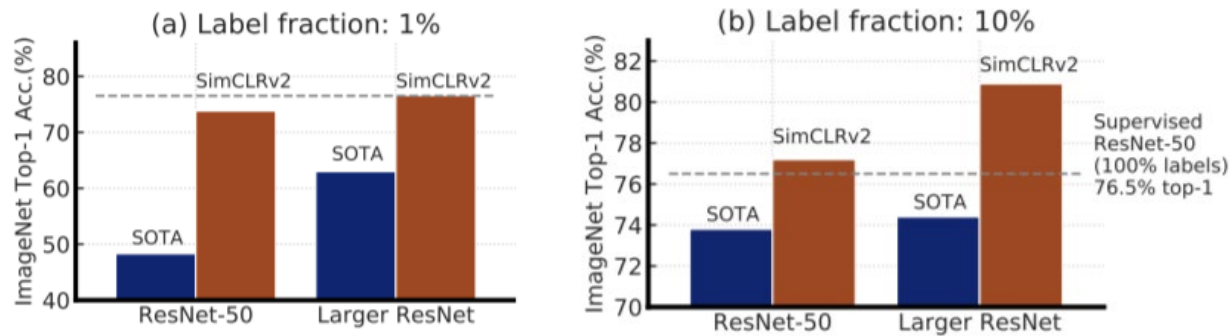
Method	Label fraction	
	1%	10%
Label only	12.3	52.0
Label + distillation loss (on labeled set)	23.6	66.2
Label + distillation loss (on labeled+unlabeled sets)	69.0	75.1
Distillation loss (on labeled+unlabeled sets)	68.9	74.3

Distillation on labeled dataset alone is not sufficient.

Using unlabeled examples largely improve distillation.

Chen et al. "Big Self-Supervised Models are Strong Semi-Supervised Learners." *NeurIPS* 2020.

SimCLR+: Improving the state of the art by a large margin



Greatly improving previous SOTA on ImageNet
(with 1% or 10% labels)

Method	Architecture	Top-1		Top-5	
		Label fraction 1%	Label fraction 10%	Label fraction 1%	Label fraction 10%
Supervised baseline [30]	ResNet-50	25.4	56.4	48.4	80.4
<i>Methods using unlabeled data in a task-specific way:</i>					
Pseudo-label [11, 30]	ResNet-50	-	-	51.6	82.4
VAT+Entropy Min. [37, 38, 30]	ResNet-50	-	-	47.0	83.4
Mean teacher [39]	ResNeXt-152	-	-	-	90.9
UDA (w. RandAug) [14]	ResNet-50	-	68.8	-	88.5
FixMatch (w. RandAug) [15]	ResNet-50	-	71.5	-	89.1
S4L (Rot+VAT+Entropy Min.) [30]	ResNet-50 (4×)	-	73.2	-	91.2
MPL (w. RandAug) [2]	ResNet-50	-	73.8	-	-
CowMix [40]	ResNet-152	-	73.9	-	91.2
<i>Methods using unlabeled data in a task-agnostic way:</i>					
InstDisc [17]	ResNet-50	-	-	39.2	77.4
BigBiGAN [41]	RevNet-50 (4×)	-	-	55.2	78.8
PIRL [42]	ResNet-50	-	-	57.2	83.8
CPC v2 [19]	ResNet-161(*)	52.7	73.1	77.9	91.2
SimCLR [1]	ResNet-50	48.3	65.6	75.5	87.8
SimCLR [1]	ResNet-50 (2×)	58.5	71.7	83.0	91.2
SimCLR [1]	ResNet-50 (4×)	63.0	74.4	85.8	92.6
BYOL [43] (concurrent work)	ResNet-50	53.2	68.8	78.4	89.0
BYOL [43] (concurrent work)	ResNet-200 (2×)	71.2	77.7	89.5	93.7
<i>Methods using unlabeled data in both ways:</i>					
SimCLRv2 distilled (ours)	ResNet-50	73.9	77.5	91.5	93.4
SimCLRv2 distilled (ours)	ResNet-50 (2×+SK)	75.9	80.2	93.0	95.0
SimCLRv2 self-distilled (ours)	ResNet-152 (3×+SK)	76.6	80.9	93.4	95.5

Chen et al. "Big Self-Supervised Models are Strong Semi-Supervised Learners." *NeurIPS* 2020.